GHXTox: Peptide toxicity prediction by fusing a task-adaptive protein language model with pLDDT-aware multi-scale 3D structural modeling

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Abstract

Peptide toxicity is a major safety concern in peptide-based drug development, motivating accurate computational screening methods. Existing methods have begun to explore multi-dimensional (multimodal) feature fusion and toxicity prediction based on the three-dimensional (3D) conformations of peptides, yet most of them still underutilize 3D structural information, and systematic multi-dimensional fusion remains rare. Here, we propose GHXTox, a hierarchical dual-level fusion framework in which a 3D structural expert first integrates sequence semantics with spatial structural features through predicted local-distance difference test (pLDDT)-aware cross-attention, and is then fused at the probability level with a task-adaptive one-dimensional (1D) sequence expert using weights frozen on training out-of-fold (OOF) predictions. The 1D branch analyzes sequences with a protein language model (ProtT5 fine-tuned with low-rank adaptation, LoRA) and a bidirectional Mamba state-space model. The 3D expert integrates an ESMFold-derived structural anchor, pLDDT-aware residue geometry, full-atom-derived chemical sites, and an availability-masked conformer-statistics residual for peptides with valid ensemble information; samples without conformer statistics fall back exactly to the single-structure pathway. On the frozen historical benchmark test sets, GHXTox achieves MCC values of 0.8726 and 0.6746 and AUPRC values of 0.9675 and 0.8043 on Test1 and Test2, respectively. Further analyses characterize residue-level sequence sensitivity, confidence–gate associations, chemical-site perturbations, and expert disagreement; these findings are interpreted as model-level evidence rather than causal biological mechanisms. Frozen-model analyses indicate that the conformer contribution is small and non-uniform: under a frozen structural expert, substantial geometric variation does not necessarily translate into stable outcome-aligned probability variation.

Keywords: peptide toxicity prediction; multimodal deep learning; protein language model; bidirectional Mamba; pLDDT-aware multi-scale 3D modeling; interpretability

1 INTRODUCTION

As biomedical research advances from the cellular to the molecular level, peptides have become increasingly important as a versatile class of bioactive molecules and are now widely used in the targeted treatment of diseases such as Parkinson's disease and cancer [1]. Peptide toxicity is a critical safety consideration throughout the drug development pipeline, and the accurate and efficient identification of toxic peptides remains a major bottleneck in peptide-based drug discovery [2]. Because conventional experimental toxicity assays are costly, time-consuming, and difficult to scale, reliable computational predictors are highly desirable.

Computational peptide toxicity prediction has evolved from traditional machine learning methods based on handcrafted physicochemical features toward deep learning approaches with stronger representation-learning capability. Current mainstream methods can be roughly grouped into two categories. The first category operates on sequence-level (one-dimensional, 1D) and/or atomic-level (two-dimensional, 2D) representations. For example, ToxMSRC combines continuous bag-of-words (CBOW) embeddings, multi-scale convolutional neural networks,

and a bidirectional long short-term memory network with residual connections, achieving strong performance on the benchmark dataset [2]. ToxPLTC integrates pooled ProtT5 embeddings, borderline-SMOTE augmentation, and a text convolutional neural network [3]. ToxiPep fuses context-aware sequence representations with atomic-level graphs through a cross-attention mechanism [4]. HyPepTox-Fuse further combines embeddings from multiple protein language models with conventional descriptors [5]. Although these models achieve strong performance within their respective modalities, they largely neglect the three-dimensional (3D) conformations of peptides, which are closely related to toxicity; cross-dimensional information interaction and mutual validation remain limited, and their interpretability is mostly confined to the sequence level.

The second category attempts to exploit 3D structural information. PeptiTox uses ESM2 sequence embeddings together with ESMFold-predicted 3D structures and geometric graph neural networks [6]. StrucToxNet feeds ProtT5 sequence embeddings and ESMFold-predicted structures into an equivariant graph neural network (EGNN) and systematically analyzes structural quality [7]. These studies demonstrate the value of 3D conformations for toxicity prediction. However, such methods are constrained by the limited accuracy of current structure prediction models on short peptides and typically rely on a single predicted conformation; structural confidence, i.e., the predicted local distance difference test (pLDDT) score, is either discarded through hard filtering or ignored, and multi-conformer ensembles are not considered. Moreover, the fusion between the sequence and structure branches remains relatively shallow, and no systematic multi-dimensional fusion that combines a task-adaptive sequence model with confidence-aware multi-scale structure modeling has been established.

To address these limitations, we propose GHXTox, an interpretable hierarchical dual-level fusion framework for peptide toxicity prediction. In the first level, a sequence-enhanced 3D structural expert integrates sequence semantics with spatial structural features through pLDDT-aware cross-attention. In the second level, this structural expert is fused with a task-adaptive 1D sequence expert at the probability level, with weights frozen on training out-of-fold (OOF) predictions. In the 1D branch, a task-adaptive protein language model (ProtT5 fine-tuned with low-rank adaptation, LoRA) is combined with a bidirectional Mamba state-space model for sequence analysis [8–10]. In the 3D branch, ESMFold provides a common single-structure anchor for all peptides [11]. AlphaFold2/ColabFold, PepFlow, and BioEmu were examined as conformer-generation references during structural development [12–15]. The conformer-residual module was trained on 2,795 peptides with validated ensemble statistics, comprising 761 PepFlow records of 11–15 residues and 2,034 AlphaFold2/ColabFold records of 16–30 residues. A three-dimensional source indicator conditions the residual head, while a hard availability mask preserves the single-structure output for all uncovered records. Frozen benchmark inference used only validated PepFlow and AlphaFold2/ColabFold conformer caches under source- and sequence-matching rules fixed independently of the benchmark labels. BioEmu was not included in the benchmark inputs. Gated sequence–structure cross-attention in the adjacent neuropeptide-prediction task further informs the interaction design, without transferring its labels or task-specific decision rule [16]. On this basis, we design a multi-scale structural information extraction strategy that progressively captures structural cues from local residue functional groups (chemical sites) to the whole peptide, and the pLDDT confidence output by structure prediction models is exploited through continuous down-weighting and dynamic gating of structural features. The 1D ensemble probability is first calibrated using training out-of-fold predictions, after which the calibrated 1D probability and the structural-expert probability are combined using OOF-frozen fusion weights. Ablation experiments show that the contribution of structural evidence is dataset- and metric-dependent: fusion improves several fixed-threshold and probability-quality measures on Test1 while maintaining broadly comparable performance on Test2. Post hoc analyses characterize residue-level sensitivity, confidence gating, chemical-site perturbations, and expert disagreement without assigning causal biological meaning to these signals.

In summary, the main contributions of this work are threefold: (1) we propose GHXTox, an interpretable hierarchical dual-level fusion framework that integrates a task-adaptive 1D protein language model with pLDDT-aware multi-scale 3D structure modeling; (2) the 3D branch designs a multi-scale structural information extraction strategy spanning from local residue functional groups to the whole peptide, and incorporates pLDDT confidence into structural feature processing through continuous down-weighting and dynamic gating; and (3) extensive experiments on the frozen historical benchmark test sets show that GHXTox improves several fixed-threshold and probability-quality measures on Test1 while maintaining broadly comparable performance on Test2, while ablation, robustness, and interpretability analyses characterize the conditions under which sequence and structural evidence are complementary.

2 MATERIALS AND METHODS

2.1 Datasets

We evaluate GHXTox on the benchmark datasets constructed by CAPTP [17] and subsequently adopted by ToxMSRC [2]. Following the data construction protocol of ToxMSRC [2], toxic peptide sequences were merged from three databases, namely CSM-Toxin [18], ToxinPred2 [19], and ATSE [20], and sequences longer than 50 residues were removed. Non-toxic sequences were retrieved from UniProt [21] using the query "NOT KW-0800 AND NOT KW-0020 AND reviewed:true", and sequences containing non-standard amino acids or exceeding 50 residues were excluded. After combining the two groups and removing samples with conflicting labels, CD-HIT [22] was applied with a sequence-identity threshold of 0.9 to eliminate redundant sequences, yielding 2,138 toxic and 5,375 non-toxic peptides. Following the same partition as ToxMSRC [2], we used 85% of the data as the training set and the remaining 15% as the benchmark Test1 set, resulting in 1,818 toxic and 4,569 non-toxic training peptides and 320 toxic and 806 non-toxic Test1 peptides. An additional benchmark Test2 set, containing 46 toxic and 536 non-toxic peptides, is used for frozen historical evaluation. Both Test1 and Test2 had been observed during historical project development and are therefore treated as frozen historical benchmarks rather than newly blinded external validation sets; their labels were not used in the formal OOF calibration, fusion-weight selection, or threshold-freezing procedure. ToxinPred 3.0 [23] was consulted as an external method and data-source reference during project development, but its records were not added to the present training set or to the formal Test1/Test2 result tables. Table 1 summarizes the statistics of the datasets used in this study.

Table 1: Statistics of the datasets used in this study.

Dataset	Toxic	Non-toxic	Total	Use
Training (benchmark)	1,818	4,569	6,387	Model training
Test1 (benchmark)	320	806	1,126	Benchmark test
Test2	46	536	582	Benchmark test

2.2 Overview of the Model

Figure 1 shows the GHXTox architecture, a hierarchical dual-level fusion framework for peptide toxicity prediction. A task-adaptive one-dimensional (1D) sequence expert and a sequence-enhanced three-dimensional (3D) structural expert generate independent toxicity probabilities. Within the structural expert, sequence semantics and spatial structure interact through pLDDT-aware cross-attention; at the top level, the two expert probabilities are fused with weights

frozen on training out-of-fold (OOF) predictions. The framework therefore combines intra-expert feature-level interaction with inter-expert probability-level fusion.

Figure 1: Overall architecture of GHXTox. GHXTox adopts a hierarchical dual-level fusion design. The task-adaptive 1D sequence expert extracts toxicity-related sequence features with a LoRA-fine-tuned ProtT5 encoder and a bidirectional Mamba network, followed by three-seed ensembling and out-of-fold (OOF) Platt calibration to produce p_{1D}. The sequence-enhanced 3D structural expert extracts multi-scale structural information through an ESM2-enhanced sequence encoder, a pLDDT-aware equivariant graph neural network, full-atom chemical-site modeling, and an availability-masked conformer residual branch. Within this expert, sequence and spatial residue representations are integrated through pLDDT-aware cross-attention to produce p_{SE3D}. The two expert probabilities are fused with frozen weights of 0.72 and 0.28, and the final toxicity decision is obtained with a threshold of 0.4702225653.

2.2.1 Task-Adaptive 1D Sequence Expert

The 1D sequence expert takes the peptide sequence as input and produces a calibrated toxicity probability. Residue-level representations from a ProtT5 protein language model [8], adapted to the toxicity task by low-rank adaptation (LoRA) [9], are processed by a bidirectional Mamba state-space encoder [10] that models dependencies in both sequence directions. After masked pooling and a linear classification head, three models trained with different random seeds are averaged, and the resulting probability is calibrated by Platt scaling [24] fitted on training OOF predictions, yielding p_{1D}. Details are given in Section 2.3.

2.2.2 Sequence-Enhanced 3D Structural Expert

The 3D structural expert organizes evidence at five scales: residue semantics, spatial residue neighborhoods, full-atom chemical sites, peptide-level properties, and conformer-ensemble statistics. An internal sequence-semantic subbranch projects amino acid types, functional-group properties, physicochemical descriptors, and ESM2 residue embeddings [11] through a two-layer Transformer [25] to produce contextualized sequence states. In parallel, a spatial-geometric subbranch constructs a hybrid residue graph with sequential and spatial-neighborhood edges and refines residue states through three pLDDT-aware equivariant graph neural network (EGNN) layers [26]. A full-atom chemical-site subbranch extracts sites for charged, aromatic, hydrophobic, donor, and acceptor groups, propagates messages over them [27], and returns the resulting features to the residue level through a zero-initialized residual. When conformer ensembles are available, statistics such as residue RMSF, pairwise distance variance, contact occupancy, conformer RMSD, and radius of gyration are combined in an availability-masked residual branch; without valid statistics, this residual is exactly zero. The contextualized sequence states then retrieve spatial evidence through pLDDT-aware cross-attention, followed by peptide-level confidence gating; after masked pooling, global physicochemical features, and MLP classification, the expert emits p_{SE3D}. Details are given in Section 2.4.

2.2.3 Hierarchical Fusion and Toxicity Decision

The calibrated 1D probability and the structural-expert probability are combined linearly in probability space with weights frozen on training OOF predictions:

$$p_{\mathrm{GHXTox}} = 0.72\, p_{1D} + 0.28\, p_{SE3D}. \tag{1}$$

The larger sequence weight is an empirical OOF-selection outcome, not a manually assigned prior; no stacking classifier or sample-specific routing is added. The fused probability is converted

into a binary prediction with the same frozen decision threshold:

$$\hat{y} = \begin{cases} 1, & p_{\mathrm{GHXTox}} \geq 0.4702225653, \\ 0, & p_{\mathrm{GHXTox}} < 0.4702225653, \end{cases} \tag{2}$$

where $\hat{y} = 1$ denotes a toxic peptide and $\hat{y} = 0$ a non-toxic peptide. The calibration parameters, fusion weights, and threshold are all determined from training OOF predictions, without using Test1 or Test2 labels, and are applied unchanged to both benchmarks (Section 2.5.4).

2.3 Task-Adaptive 1D Sequence Modeling

The 1D sequence expert is designed to learn toxicity-related contextual semantics and sequential dependencies directly from peptide amino acid sequences. Instead of reducing a frozen protein-language-model output to a fixed mean vector, GHXTox preserves the residue-level representations generated by ProtT5 and adapts the pretrained encoder to the peptide toxicity prediction task through low-rank adaptation (LoRA). The resulting residue-level representations are then processed by a bidirectional Mamba state-space encoder to jointly model local residue patterns, N-to-C dependencies, and C-to-N dependencies. The encoded residue features are aggregated via masked mean pooling and passed to a linear classification head to produce the single-model toxicity probability. Finally, three models trained on the full training set with different random seeds are combined by equal-weight probability averaging, followed by Platt calibration fitted exclusively on training out-of-fold (OOF) predictions, yielding the final 1D expert probability p_{1D}.

This design organizes pretrained protein semantics, parameter-efficient task adaptation, bidirectional sequence modeling, and probability calibration into a coherent processing pipeline. While most of the pretrained knowledge in ProtT5 is preserved, LoRA enables task-specific adaptation of the residue representations. The bidirectional Mamba encoder further reorganizes these representations according to toxicity-related sequential patterns, whereas multi-seed ensembling and out-of-fold calibration reduce training variance and align the calibrated probability scale with that of the downstream structural expert.

2.3.1 Residue Tokenization and LoRA-Adapted ProtT5

Given a peptide sequence of length L, we first convert its amino acid residues into the space-separated format required by the ProtT5 tokenizer and map them to token indices. Special tokens are inserted automatically, and sequences of different lengths are padded to a uniform length within each batch; the maximum token length is 128. Under the current tokenizer configuration, the first L token positions correspond to the L amino acid residues, followed by the EOS token and, when required, padding tokens. A binary residue-validity mask m_i is therefore used, with $m_i = 1$ for valid residue positions and $m_i = 0$ otherwise; the same mask is applied in the Mamba scans and the subsequent pooling.

The tokenized sequence is processed by the ProtT5-XL-UniRef50 encoder [8] with LoRA, producing residue-level representations $\mathbf{H}^{\mathrm{PLM}} \in \mathbb{R}^{L \times 1024}$. Keeping the complete residue-level representation matrix enables the downstream encoder to preserve position-specific contextual semantics; each residue vector encodes not only its amino acid identity but also its conditional context within the complete peptide sequence.

To adapt the general-purpose ProtT5 encoder to the peptide toxicity prediction task, we apply low-rank adaptation (LoRA) [9] to the linear transformations within the encoder. For an original linear layer $\mathbf{y} = \mathbf{W}_0\mathbf{x} + \mathbf{b}$, the pretrained weight matrix $\mathbf{W}_0$ is frozen, and the task-specific update is factorized into two trainable low-rank matrices:

$$\mathbf{W} = \mathbf{W}_0 + \frac{\alpha}{r}\mathbf{B}\mathbf{A}, \qquad \mathbf{A} \in \mathbb{R}^{r \times d_{\mathrm{in}}}, \quad \mathbf{B} \in \mathbb{R}^{d_{\mathrm{out}} \times r}. \tag{3}$$

We set the LoRA rank to $r = 8$, the scaling coefficient to $\alpha = 16$, and the LoRA dropout rate to 0.1, yielding a scaling factor of $\alpha/r = 2$. Matrix $\mathbf{A}$ is initialized with Kaiming initialization, whereas $\mathbf{B}$ is initialized to zero; consequently, the LoRA branch initially leaves the pretrained transformation unchanged and progressively learns toxicity-specific updates during training. Unlike conventional configurations that apply LoRA only to selected attention projections, the current implementation augments all linear layers within the ProtT5 encoder; the pretrained base transformations and embeddings remain frozen, whereas the low-rank parameters and the downstream sequence-classification modules are trainable.

The 1024-dimensional ProtT5 representations are then projected into the 192-dimensional hidden space used by the Mamba encoder [10]:

$$\mathbf{X}^{(0)} = \mathrm{Dropout}\left(\mathrm{GELU}\left[\mathrm{LN}\left(\mathbf{H}^{\mathrm{PLM}}\mathbf{W}_p + \mathbf{b}_p\right)\right]\right), \qquad \mathbf{X}^{(0)} \in \mathbb{R}^{L\times 192}.$$

where LN denotes layer normalization. The projection block consists of a linear transformation, layer normalization, GELU activation, and dropout, reducing the computational dimension for the downstream state-space encoder while providing a task-specific nonlinear transformation of the ProtT5 residue representations.

2.3.2 Bidirectional Mamba Sequence Encoder

Although ProtT5 provides contextualized residue representations, these representations primarily encode general protein semantics learned during large-scale pretraining. To further model the sequential dependencies specifically associated with peptide toxicity, two bidirectional Mamba layers are placed after the ProtT5 projection block. Each directional block follows the standard selective state-space formulation of Mamba [10]: the input is projected into content and gating branches, processed by a depthwise convolution, discretized with input-dependent step sizes, and updated through a recurrent state of dimension 16. In compact form, the discretized transition and the state update are

$$\bar{\mathbf{A}}_t = \exp(\Delta_t \mathbf{A}), \qquad \mathbf{S}_t = \bar{\mathbf{A}}_t \odot \mathbf{S}_{t-1} + \bar{\mathbf{B}}_t \odot \widetilde{\mathbf{U}}^{\mathrm{m}}_t, \tag{4}$$

followed by a gated output projection. At padding positions, the convolved feature and the step size are zeroed, preventing padding tokens from injecting information into the recurrent states.

A unidirectional scan accesses only one side of the sequence context, whereas peptide toxicity may depend on residue patterns associated with both termini and with interactions spanning different sequence positions. In the ℓ-th bidirectional layer, forward and backward scans are therefore applied to the normalized sequence using the residue-validity mask $\mathbf{m}$, and the two directional outputs are concatenated and fused by a linear layer with a residual connection:

$$\begin{aligned}
\mathbf{F}^{(\ell)} &= \mathcal{M}_f^{(\ell)}(\mathrm{LN}(\mathbf{X}^{(\ell)}), \mathbf{m}), \\
\mathbf{R}^{(\ell)} &= \mathrm{Reverse}\left[\mathcal{M}_b^{(\ell)}\left(\mathrm{Reverse}\left[\mathrm{LN}(\mathbf{X}^{(\ell)})\right], \mathrm{Reverse}(\mathbf{m})\right)\right], \\
\mathbf{X}^{(\ell+1)} &= \mathbf{X}^{(\ell)} + \mathrm{Dropout}\left(\mathrm{GELU}\left(\mathbf{W}_m\left[\mathbf{F}^{(\ell)} \parallel \mathbf{R}^{(\ell)}\right] + \mathbf{b}_m\right)\right),
\end{aligned} \tag{5}$$

where $\parallel$ denotes feature concatenation and the backward-scan output is restored to the original residue order. Two such layers are stacked in the final model, allowing each residue position to receive conditional information from both its N-terminal and C-terminal contexts.

2.3.3 Masked Pooling and Class-Weighted Classification

After two bidirectional Mamba layers, the model produces residue-level representations $\mathbf{X}^{(2)} \in \mathbb{R}^{L\times 192}$. Because peptide sequences have variable lengths, GHXTox does not rely on a

fixed-position classification token. Instead, masked mean pooling is applied over the valid amino acid positions:

$$\mathbf{h}_{1D} = \frac{\sum_{i=1}^{L} m_i \mathbf{x}_i}{\sum_{i=1}^{L} m_i}, \tag{6}$$

where the denominator equals the number of valid residue positions. This operation converts the variable-length residue sequence into a fixed 192-dimensional peptide-level representation while ensuring that EOS and padding positions do not contribute to the pooled feature; unlike global max pooling, it aggregates distributed evidence across the entire peptide.

The pooled representation is normalized and passed through a single-output linear classification head, $z_{1D} = \mathbf{w}_c^\top \mathrm{LN}(\mathbf{h}_{1D}) + b_c$, giving the uncalibrated toxicity probability $p_{1D}^{\mathrm{raw}} = \sigma(z_{1D})$. The classification head therefore consists of layer normalization followed by a single linear layer, rather than a multilayer perceptron.

Because non-toxic peptides substantially outnumber toxic peptides in the training data, a positive-class-weighted binary cross-entropy loss is used in the current 1D expert instead of SMOTE or Borderline-SMOTE. For a training partition containing N_+ toxic and N_- non-toxic peptides, the positive-class weight is $w_+ = N_-/N_+$, and the class-weighted objective is

$$\mathcal{L}_{\mathrm{WBCE}} = -\frac{1}{N} \sum_{j=1}^{N} \left[w_+ y_j \log \sigma(z_j) + (1 - y_j) \log(1 - \sigma(z_j)) \right]. \tag{7}$$

During five-fold out-of-fold training, w_+ is calculated independently from the class counts of each fold's training partition; for full-data training, it is recalculated using the complete training set. This strategy increases the contribution of the minority toxic class without generating synthetic peptide sequences or feature vectors, while preventing any information from the held-out validation fold from entering the training procedure.

2.3.4 Three-Seed Ensemble and OOF Platt Calibration

The predictions of deep neural networks may vary because of parameter initialization, mini-batch ordering, and stochastic optimization. To reduce the variance associated with an individual training run, three 1D expert models are trained under identical data, architecture, and optimization settings using the random seeds 42, 123, and 2025. The final ensemble consists of three models trained on the complete training set, rather than the fold-specific checkpoints produced during cross-validation.

For sample j, the toxicity probability produced by the model trained with seed k is denoted by $p_j^{(k)} = \sigma(z_j^{(k)})$, $k \in \{1, 2, 3\}$, and the uncalibrated ensemble score is obtained by equal-weight arithmetic averaging:

$$s_j = \frac{1}{3} \sum_{k=1}^{3} p_j^{(k)}. \tag{8}$$

No individual seed is selected according to its performance on Test1 or Test2, and no seed-specific ensemble weight is introduced.

Although probability averaging reduces prediction variance, neural-network probabilities may remain systematically overconfident or underconfident. Platt scaling [24] is therefore applied after the three-model probability average. Training OOF predictions are first generated using the fixed group-aware five-fold partition; each training peptide receives an OOF probability from a model trained without the fold containing that peptide, and the OOF probabilities from the three random seeds are averaged to obtain the OOF ensemble score s_j^{OOF}. A two-parameter logistic calibrator is fitted to these OOF scores:

$$p_{1D,j} = \sigma\left(a \,\mathrm{logit}\left(\widetilde{s}_j\right) + b\right), \tag{9}$$

where $\mathrm{logit}(s) = \ln(s/(1-s))$, $\widetilde{s}_j = \mathrm{clip}(s_j, \varepsilon, 1-\varepsilon)$ with $\varepsilon = 10^{-6}$, and a and b denote the Platt calibration parameters; the clip prevents infinite logits for probabilities of exactly 0 or 1, and the same clipping is applied during fitting. The parameters are estimated by minimizing binary cross-entropy on the OOF scores, and the scale is reparameterized as $a = \mathrm{clip}(\exp(\eta),\, 0.02, 50) > 0$, which guarantees a monotonic transformation and therefore preserves the sample ranking of the ensemble score. For benchmark test or previously unseen peptides, the frozen calibrator is applied to the three-seed average: $p_{1\mathrm{D}} = \sigma\left(a\,\mathrm{logit}\left(\mathrm{clip}\left(\frac{1}{3}\sum_{k=1}^{3} p^{(k)},\ \varepsilon,\ 1-\varepsilon\right)\right) + b\right)$. The labels of Test1 and Test2 are not used for ensemble-member selection, calibration fitting, or 1D threshold optimization. The calibrated probability $p_{1\mathrm{D}}$ is the sequence-expert output used in the top-level probability fusion described in Section 2.2.3; no additional Platt calibrator is fitted after the final GHXTox probability fusion.

2.4 pLDDT-Aware Multi-Scale 3D Structural Modeling

Peptide toxicity is influenced not only by amino acid composition and linear residue order but also by the three-dimensional (3D) arrangement of charged groups, hydrophobic regions, aromatic systems, and sulfur-containing moieties. However, peptides are generally short, conformationally flexible, and poorly covered by experimentally determined structures. Treating a single predicted conformation as an error-free structural observation may therefore amplify structure-prediction errors. Conversely, representing a peptide with only Cα coordinates or peptide-level summary descriptors fails to capture the local spatial-chemical environments that are relevant to toxicity.

To address these limitations, GHXTox employs a pLDDT-aware multi-scale structural expert that captures complementary information at five levels: residue semantics, spatial residue neighborhoods, full-atom chemical sites, peptide-level global properties, and conformer-ensemble statistics. The model first extracts residue-level Cα coordinates, full-atom coordinates, and normalized pLDDT confidence scores from the predicted structure. An internal sequence-semantic subbranch combines amino acid residue types, functional-group categories, explicit physicochemical descriptors, and ESM2 residue embeddings to generate contextualized sequence states. In parallel, a spatial-geometric subbranch constructs a hybrid residue graph containing both sequential adjacency edges and spatial-neighborhood edges and propagates geometric information through pLDDT-aware equivariant graph layers.

To overcome the limited chemical resolution of a Cα-only residue graph, GHXTox additionally extracts typed chemical sites from the full-atom structure and performs message passing using dual-scale distances, directional relationships, interaction-compatibility channels, and structural confidence. For peptides with available conformer ensembles, an additional statistical residual branch summarizes residue fluctuations, distance variation, and contact occupancy. The residual is constrained to exactly zero when conformer information is unavailable.

Rather than discarding low-confidence structures through hard filtering, GHXTox treats pLDDT as a continuous reliability signal and incorporates it into geometric edge weighting, chemical-site messages, residue-level structural gating, and peptide-level structural gating. Low-confidence structural evidence is therefore down-weighted rather than removed, preserving structural coverage while limiting the influence of uncertain coordinates. pLDDT is interpreted here as a local confidence estimate produced by the structure-prediction model, not as an experimentally determined measure of structural accuracy, conformational flexibility, or biological activity. The detailed architecture of the structural expert is illustrated in Figure 2.

2.4.1 Peptide Structure Prediction and Conformer Preparation

For a peptide sequence of length L, $S = (a_1, a_2, \ldots, a_L)$, ESMFold is first used to predict its three-dimensional structure [11]. The predicted structure is stored as a full-atom PDB file, from which residue-level Cα coordinates, backbone coordinates, full-atom-derived chemical-site

Figure 2: Detailed architecture of the sequence-enhanced structural expert. The structural expert integrates ESM2-enhanced residue semantics, a pLDDT-aware geometric residue graph, full-atom chemical-site message passing, and availability-masked conformer-ensemble statistics. Residue-level structural states are continuously modulated by node-level confidence before serving as keys and values in sequence–structure cross-attention. A peptide-level confidence gate further controls the fused structural context, while conformer statistics provide a zero-initialized logit residual only when valid ensemble information is available.

coordinates, and pLDDT scores are extracted. This operation is denoted

$$(\mathbf{X}, \mathcal{A}, \mathbf{q}) = \mathrm{ESMFold}(\mathcal{S}), \qquad \mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_L]^\top \in \mathbb{R}^{L\times 3}, \quad \mathbf{q} = (q_1, \ldots, q_L) \in [0,1]^L,$$

where $\mathbf{X}$ contains the Cα coordinates, $\mathcal{A}$ denotes the full-atom-PDB-derived backbone and chemical-site geometry, and $\mathbf{q}$ contains the residue-level pLDDT confidence scores rescaled to $[0, 1]$. The Cα coordinates serve as the primary spatial variables of the residue-level equivariant graph, whereas the full-atom coordinates are used to construct local chemical sites. Before geometric encoding, the coordinates are centered at the peptide centroid to remove the effect of absolute translation.

A single predicted structure provides a common structural anchor but cannot represent potential conformational heterogeneity. The conformer-residual stage was therefore trained on a source-aware full2795 cohort containing 2,795 peptides with up to eight validated, aligned conformers per peptide. PepFlow supplies 761 training records of 11–15 residues, whereas AlphaFold2/ColabFold supplies 2,034 records of 16–30 residues [12–14]. These sources are represented explicitly in the residual head; nevertheless, generator identity remains partially coupled to peptide length and is treated as a limitation rather than as an independently identified causal factor.

For frozen benchmark inference, conformer statistics were admitted only from a validated cache whose source and sequence-matching policies were fixed without using Test1 or Test2 labels. The Test1 cache contains eight conformers for each of 486 peptides: 130 PepFlow-derived records of 3–15 residues and 356 AlphaFold2/ColabFold-derived records of 16–30 residues. Test2 reuses 284 of these ensembles through an exact unique-sequence join; no independent Test2 conformer generation or label-dependent cache selection is performed. For every admitted ensemble, residue indices are matched to a common sequence order. An ensemble is represented as $\mathcal{C} = \{\mathbf{X}^{(1)}, \ldots, \mathbf{X}^{(M)}\}$, $1 \le M \le 8$, where $\mathbf{X}^{(m)} \in \mathbb{R}^{L\times 3}$ denotes the Cα coordinates of conformer m. Rigid-body alignment is performed before the ensemble statistics are calculated, ensuring that the resulting variation reflects internal geometric changes rather than arbitrary global rotations or translations.

Because conformer ensembles are not available for every training peptide, a binary availability indicator is defined as

$$c_{\mathrm{conf}} = \begin{cases} 1, & \text{if valid ensemble statistics are available,} \\ 0, & \text{otherwise.} \end{cases} \tag{10}$$

When $c_{\mathrm{conf}} = 0$, the conformer-statistics tensors are filled with zeros, and the corresponding residual contribution is explicitly suppressed by the availability indicator. Missing ensemble information is therefore not interpreted as evidence of structural rigidity; instead, the model falls back exactly to the single-structure pathway. Under the benchmark cache, valid conformer statistics are available for 486/1,126 (43.16%) Test1 peptides and 284/582 (48.80%) Test2 peptides. Only these covered records may receive a conformer residual; all remaining records keep the exact single-structure prediction.

The generated conformers are computational samples rather than experimentally determined trajectories or rigorous thermodynamic ensembles. Consequently, RMSF, contact occu-

pancy, and distance variance are interpreted as sampling-derived descriptors of conformational variation rather than direct measurements of peptide dynamics in solution.

2.4.2 Residue Feature Construction and Hybrid Geometric Graph

The structural expert constructs both an internal sequence representation and a spatial-geometric representation at the residue level. The internal sequence-semantic subbranch described here uses frozen ESM2 residue embeddings and a lightweight Transformer. It is distinct from the LoRA-ProtT5 and bidirectional Mamba expert described in Section 2.3. The former supports residue-level sequence–structure alignment inside the structural expert, whereas the latter acts as an independent top-level 1D prediction expert.

For residue i, the input to the internal sequence-semantic subbranch is

$$\mathbf{u}_i^{\mathrm{seq}} = \left[\mathbf{e}_i^{\mathrm{AA}} \parallel \mathbf{e}_i^{\mathrm{group}} \parallel \mathbf{f}_i^{\mathrm{res}} \parallel P_{\mathrm{ESM}}\left(\mathbf{e}_i^{\mathrm{ESM2}}\right)\right], \tag{11}$$

where $\mathbf{e}_i^{\mathrm{AA}} \in \mathbb{R}^{32}$ is the amino acid category embedding, $\mathbf{e}_i^{\mathrm{group}} \in \mathbb{R}^{12}$ is the functional-group category embedding, $\mathbf{f}_i^{\mathrm{res}} \in \mathbb{R}^{42}$ denotes the explicit residue-level physicochemical descriptors, and $\mathbf{e}_i^{\mathrm{ESM2}} \in \mathbb{R}^{1280}$ is the ESM2-650M residue representation; the projection P_{ESM} maps the ESM2 representation to 128 dimensions. Stacking the residue-level feature vectors yields the sequence-feature matrix $U_{\mathrm{seq}} \in \mathbb{R}^{L\times 214}$, where $214 = 32+12+42+128$. After a linear projection and positional encoding, a two-layer Transformer encoder with hidden dimension 128, four attention heads, and a feed-forward dimension of 512 produces the contextualized sequence states $\mathbf{H}^{\mathrm{seq}} \in \mathbb{R}^{L\times 128}$. This representation encodes the contextual sequence semantics of each residue and subsequently serves as the query representation in the sequence–structure cross-attention module.

In the spatial subbranch, the initial representation of residue i is

$$\mathbf{h}_i^{(0)} = P_{\mathrm{sp}}\left[\mathbf{f}_i^{\mathrm{res}} \parallel \mathbf{g}_i^{\mathrm{geo}} \parallel q_i \parallel P_{\mathrm{ESM}}\left(\mathbf{e}_i^{\mathrm{ESM2}}\right)\right],$$

where $\mathbf{g}_i^{\mathrm{geo}} \in \mathbb{R}^{16}$ is a lightweight geometric descriptor derived from the Cα coordinates and pLDDT scores. The 16-dimensional descriptor includes (i) residue-level pLDDT and local confidence statistics; (ii) radial position, contact fractions, and soft spatial density; (iii) nearest-neighbor, nonlocal-neighbor, and local sequential distances; and (iv) sine/cosine encodings of the Cα-trace angles and torsions. These explicit descriptors provide stable local geometric priors without introducing an overly complex handcrafted representation for short and uncertain predicted peptide structures.

GHXTox then constructs a hybrid residue graph:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}_{\mathrm{seq}} \cup \mathcal{E}_{\mathrm{knn}}), \tag{12}$$

where each residue corresponds to one node. Sequential edges $\mathcal{E}_{\mathrm{seq}}$ connect adjacent residues and preserve peptide-chain continuity, whereas $\mathcal{E}_{\mathrm{knn}}$ connects each residue to its 10 nearest spatial neighbors and is symmetrized. The hybrid graph therefore captures both covalent sequence order and nonlocal spatial contacts while avoiding large variations in graph density that may arise from a single fixed-distance cutoff. Equivalently, $A_{ij} \in \{0,1\}$ denotes whether residues i and j are connected by a sequential or k-nearest-neighbor edge in the hybrid graph.

For residues i and j, the Cα distance $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|_2$ is encoded by 16 radial basis functions with centers μ_k uniformly spaced over 0–24 Å, $\phi_k(d_{ij}) = \exp[-\gamma(d_{ij} - \mu_k)^2]$ with $\gamma = 1/(\mu_{k+1} - \mu_k)^2$. The base edge representation additionally includes the pairwise structural confidence and the normalized sequence separation:

$$c_{ij} = \sqrt{q_i q_j}, \qquad s_{ij} = \frac{\log(1 + |i-j|)}{\log(L+1)}. \tag{13}$$

Thus, sequentially adjacent residues receive small values, whereas residues separated by longer sequence intervals approach one. The resulting edge feature is therefore $\mathbf{e}_{ij} = [\boldsymbol{\phi}(d_{ij}) \parallel c_{ij} \parallel s_{ij}]$. This formulation allows the model to distinguish sequentially adjacent contacts, long-range spatial contacts, and contacts supported by different levels of structural confidence.

2.4.3 pLDDT-Aware Equivariant Graph Encoding

To learn three-dimensional representations without relying on a particular global coordinate frame, the spatial branch employs three E(3)-equivariant graph neural network layers [26]. The network updates residue features using rotation- and translation-invariant quantities while updating coordinates equivariantly. Consequently, a global rigid transformation of the peptide structure does not change the predicted toxicity, whereas intermediate coordinate representations transform consistently with the input.

At layer ℓ, the message from residue j to residue i is computed as $\mathbf{m}_{ij}^{(\ell)} = \phi_e^{(\ell)}[\mathbf{h}_i^{(\ell)} \parallel \mathbf{h}_j^{(\ell)} \parallel \mathbf{e}_{ij}^{(\ell)}]$, where $\phi_e^{(\ell)}$ is a message multilayer perceptron. To prevent uncertain or distant contacts from contributing as strongly as local high-confidence contacts, a continuous edge weight is defined as

$$w_{ij}^{(\ell)} = A_{ij} \, \exp\left(-\frac{d_{ij}^{(\ell)}}{12}\right) \left[q_{\min} + (1 - q_{\min}) \, c_{ij}^{\rho}\right], \qquad q_{\min} = 0.1, \quad \rho = 1.5, \tag{14}$$

The term $\exp(-d_{ij}^{(\ell)}/12)$ imposes a smooth distance decay with a characteristic decay length of 12 Å, and both the distance and the edge weight are recomputed at each layer from the current coordinates. Here, $A_{ij} \in \{0, 1\}$ is the adjacency indicator of the hybrid graph defined in Section 2.4.2, and c_{ij} is the pairwise structural confidence defined in Section 2.4.2. The minimum weight $q_{\min}$ prevents low-confidence regions from being completely disconnected, whereas the exponent ρ increases the relative contribution of high-confidence contacts. The distance-decay term further reduces the influence of more distant spatial neighbors.

Messages received by node i are aggregated with a normalization floor, and the EGNN layer updates node features and coordinates as

$$\begin{aligned}
\mathbf{m}_i^{(\ell)} &= \frac{\sum_j w_{ij}^{(\ell)} \, \mathbf{m}_{ij}^{(\ell)}}{\max\left(\sum_j w_{ij}^{(\ell)}, \, 1\right)}, \\
\mathbf{h}_i^{(\ell+1)} &= \mathrm{LN}\left\{\mathbf{h}_i^{(\ell)} + \phi_h^{(\ell)}\left[\mathbf{h}_i^{(\ell)} \parallel \mathbf{m}_i^{(\ell)}\right]\right\}, \\
\mathbf{x}_i^{(\ell+1)} &= \mathbf{x}_i^{(\ell)} + 0.05 \, \frac{\sum_j \alpha_{ij}^{(\ell)} \left(\mathbf{x}_i^{(\ell)} - \mathbf{x}_j^{(\ell)}\right)}{\max\left(\sum_j w_{ij}^{(\ell)}, \, 1\right)}.
\end{aligned} \tag{15}$$

where $\alpha_{ij}^{(\ell)} = \tanh[\phi_x^{(\ell)}(\mathbf{m}_{ij}^{(\ell)})] \, w_{ij}^{(\ell)}$ is a bounded coordinate-update coefficient generated from the edge message and pairwise relative vectors. The coordinates are recentered after every layer to reduce numerical drift. Because the update depends only on scalar messages, distances, and relative coordinate vectors, the operation remains equivariant to rigid rotations and translations.

Three EGNN layers are stacked to progressively enlarge the spatial receptive field. The first layer primarily encodes direct geometric neighbors, whereas subsequent layers integrate higher-order spatial environments through repeated message passing. Unlike approaches that simply append pLDDT as an ordinary node feature, GHXTox uses pLDDT to directly modulate the strength of geometric messages, allowing structural uncertainty to influence representation learning at the propagation stage.

Continuous down-weighting does not imply that a low-pLDDT residue is biologically unimportant. Low confidence may arise from prediction error or potential conformational flexibility;

the model reduces the reliability assigned to uncertain coordinate evidence while preserving the residue's sequence-semantic and physicochemical features.

2.4.4 Full-Atom Chemical-Site Message Passing

A Cα-based residue graph describes the global fold and spatial neighborhoods but cannot distinguish chemically distinct regions within the same side chain. For example, tyrosine contains both an aromatic ring and a hydroxyl group, whereas tryptophan contains an aromatic system and an indole nitrogen. Compressing each side chain into a single centroid would mix these chemically different moieties. GHXTox therefore extracts chemistry-aware pseudo-sites from the predicted full-atom PDB structure and processes them with a dedicated local chemical branch.

Up to two chemical sites are kept for each residue. Site identity is represented by an eight-dimensional multi-hot vector $\mathbf{t}_{ir} \in \{0,1\}^8$ allowing overlapping types: positive, negative, donor, acceptor, aromatic, hydrophobic, sulfur, and conformational, where r indexes the chemical site within residue i. Multiple types may be assigned to the same site. For example, the cysteine sulfur site is labeled as donor, hydrophobic, and sulfur-containing, whereas the proline ring is labeled as hydrophobic and conformationally constrained. The aromatic ring and hydroxyl group of tyrosine are represented as separate sites, as are the aromatic system and the indole NE1 nitrogen (donor) of tryptophan.

A site coordinate is obtained from the mean position of the heavy atoms defining the corresponding chemical moiety. Site orientation is represented by either a bond direction or an unsigned plane normal. Because the sign of a plane normal is arbitrary, downstream directional features use absolute cosines, preserving rotation invariance and eliminating orientation-sign ambiguity.

The initial representation of site r in residue i is

$$\mathbf{z}_{ir}^{(0)} = \phi_{\text{site}}\left[\mathbf{t}_{ir} \parallel q_i \parallel o_{ir}^{\text{valid}}\right],$$

where q_i is the pLDDT score of the parent residue and o_{ir}^{valid} indicates whether a reliable orientation could be constructed; the input is projected into a 64-dimensional hidden space.

Chemical-site edges are established between valid sites separated by no more than 16 Å. In the current configuration, same-residue site edges are kept, preserving spatial relationships between chemically different moieties in complex side chains. Each edge contains four complementary feature families. First, the raw distance in Å is encoded using 16 radial basis functions, $\boldsymbol{\phi}_{\text{raw}}(d_{uv}) \in \mathbb{R}^{16}$. Second, a peptide-size-normalized distance is constructed. The radius of gyration of the Cα coordinates is

$$R_g = \sqrt{\frac{1}{L}\sum_{i=1}^{L} \|\mathbf{x}_i - \bar{\mathbf{x}}\|_2^2}, \qquad \bar{\mathbf{x}} = \frac{1}{L}\sum_{i=1}^{L}\mathbf{x}_i, \qquad \bar{d}_{uv} = \frac{d_{uv}}{R_g}, \tag{16}$$

where $\bar{\mathbf{x}}$ denotes the centroid of the valid Cα coordinates, and the normalized distance is encoded by eight radial basis functions over the interval 0–4, $\boldsymbol{\phi}_{\text{norm}}(\bar{d}_{uv}) \in \mathbb{R}^8$. The raw representation describes the physical separation in Å, whereas the normalized representation describes the separation relative to the overall peptide size. Their combination allows the model to distinguish, for example, an 8 Å contact in a compact short peptide from an 8 Å contact in a larger peptide.

Third, four rotation-invariant orientation features are used:

$$\mathbf{o}_{uv} = \left[|\mathbf{o}_u \cdot \mathbf{r}_{uv}|,\ |\mathbf{o}_v \cdot \mathbf{r}_{vu}|,\ |\mathbf{o}_u \cdot \mathbf{o}_v|,\ o_{uv}^{\text{valid}}\right], \tag{17}$$

which describe the alignment of the source orientation with the intersite direction, the alignment of the target orientation with the reverse direction, the mutual orientation of the two

sites, and orientation availability. Fourth, six explicit compatibility channels are used: opposite-charge, donor–acceptor, aromatic–aromatic, positive–aromatic (cation–π), sulfur–sulfur, and hydrophobic–hydrophobic compatibility. These channels encode candidate chemical compatibility patterns available to the model; they do not imply that a true salt bridge, hydrogen bond, cation–π interaction, disulfide bond, or hydrophobic core is present in the predicted structure.

The full chemical-site edge representation is

$$\mathbf{e}_{uv}^{\mathrm{chem}} = \left[\boldsymbol{\phi}_{\mathrm{raw}}(d_{uv}) \parallel \boldsymbol{\phi}_{\mathrm{norm}}(\bar{d}_{uv}) \parallel \mathbf{o}_{uv} \parallel \mathbf{c}_{uv}^{\mathrm{chem}} \parallel \sqrt{q_u q_v}\right]. \tag{18}$$

At layer l, each site message is computed from the two site states and the edge representation, confidence-weighted, and aggregated over the neighborhood. After two message-passing layers, valid sites belonging to the same residue are mean-pooled, and the residue-level chemical residual and the chemical-enhanced spatial state are

$$\begin{aligned}
\mathbf{m}_{uv}^{(l)} &= \phi_{\mathrm{chem}}^{(l)}\left[\mathbf{z}_u^{(l)} \parallel \mathbf{z}_v^{(l)} \parallel \mathbf{e}_{uv}^{\mathrm{chem}}\right], \\
\bar{\mathbf{m}}_v^{(l)} &= \frac{\sum_{u\in\mathcal{N}(v)} \sqrt{q_u q_v}\, \mathbf{m}_{uv}^{(l)}}{\max\left(\sum_{u\in\mathcal{N}(v)} \sqrt{q_u q_v},\, 1\right)}, \\
\Delta\mathbf{h}_i^{\mathrm{chem}} &= \mathbf{W}_{\mathrm{chem}} \operatorname*{Mean}_{r\in\mathcal{S}_i}\left(\mathbf{z}_{ir}^{(2)}\right), \\
\mathbf{h}_i^{\mathrm{sp}} &= \mathbf{h}_i^{\mathrm{EGNN}} + \Delta\mathbf{h}_i^{\mathrm{chem}},
\end{aligned} \tag{19}$$

where the projection $\mathbf{W}_{\mathrm{chem}}$ and its bias $\mathbf{b}_{\mathrm{chem}}$ are zero-initialized, so that $\Delta\mathbf{h}_i^{\mathrm{chem}} = 0$ at the beginning of training. This conservative residual design allows the new chemical-site branch to progressively learn useful corrections from an intact geometric backbone rather than perturbing the existing spatial representation with a randomly initialized contribution.

2.4.5 Availability-Masked Conformer-Ensemble Residual

A single predicted structure provides a deterministic spatial anchor, whereas a peptide may adopt multiple related conformations. Running the complete geometric network independently on every conformer would be computationally expensive and could allow the model to exploit differences between conformer generators or uneven ensemble coverage. The current GHXTox model therefore uses a lightweight conformer-statistics residual. The single-structure expert remains the base predictor, and ensemble statistics are permitted to correct its logit only when valid conformer information is available.

For a peptide with M aligned conformers, we compute the residue-level root-mean-square fluctuation, the per-pair mean distance and distance standard deviation, and the contact occupancy at cutoff $r \in \{6, 8, 10\}$ Å:

$$\begin{aligned}
\mathrm{RMSF}_i &= \sqrt{\frac{1}{M}\sum_{m=1}^{M}\left\|\mathbf{x}_i^{(m)} - \bar{\mathbf{x}}_i\right\|_2^2}, \qquad \bar{d}_{ij} = \frac{1}{M}\sum_{m=1}^{M} d_{ij}^{(m)}, \\
\sigma_{ij}^{d} &= \sqrt{\frac{1}{M}\sum_{m=1}^{M}\left(d_{ij}^{(m)} - \bar{d}_{ij}\right)^2}, \quad O_{ij}^{(r)} = \frac{1}{M}\sum_{m=1}^{M}\mathbb{I}\left(d_{ij}^{(m)} < r\right),
\end{aligned} \tag{20}$$

with $\bar{\mathbf{x}}_i$ the mean conformer coordinate of residue i. The 8 Å contact-occupancy uncertainty $U_{ij}^{(8)} = O_{ij}^{(8)}(1 - O_{ij}^{(8)})$ is also used: values close to zero or one indicate contacts that are consistently absent or present across the sampled conformers, whereas values near 0.5 indicate contact switching.

The ensemble statistics are organized into an eight-dimensional residue-level descriptor $\mathbf{r}_i^{\mathrm{conf}} \in \mathbb{R}^8$, containing the residue RMSF; the mean pairwise distance to other residues; the

standard deviation of the mean pairwise distances associated with the residue; the mean pairwise-distance standard deviation; the mean contact occupancies at 6, 8, and 10 Å; and the mean 8 Å contact-occupancy uncertainty, which together form the eight-dimensional descriptor. A twelve-dimensional peptide-level descriptor $\mathbf{g}^{\mathrm{conf}} \in \mathbb{R}^{12}$ is also constructed, containing the normalized number of conformers; the mean, standard deviation, and maximum of RMSF; the mean, standard deviation, and maximum of the conformer RMSD; the mean and standard deviation of the radius of gyration; the mean and maximum of the pair-distance standard deviation; and the global mean of $U_{ij}^{(8)}$, which together form the twelve-dimensional descriptor. Before neural encoding, distance- and fluctuation-based statistics are rescaled by fixed constants to place heterogeneous descriptors on comparable numerical ranges; the exact scaling factors are reported in the implementation-details table.

The residue-level ensemble features are encoded and aggregated using both masked mean and masked maximum pooling, whereas the global descriptor is processed by a separate encoder, giving the ensemble embeddings $\mathbf{h}_{\mathrm{conf}}^{\mathrm{res}}$ and $\mathbf{h}_{\mathrm{conf}}^{\mathrm{global}}$. The generator is encoded by a three-dimensional one-hot vector $\mathbf{s}_{\mathrm{conf}} \in \mathbb{R}^3$ for PepFlow, AlphaFold2/ColabFold, or another admitted generator. A learned source embedding $\phi_s(\mathbf{s}_{\mathrm{conf}})$ is concatenated with the base structural embedding and the two ensemble embeddings to produce a raw logit residual

$$\Delta z_{\mathrm{conf}}^{\mathrm{raw}} = \phi_{\mathrm{conf}}\left[\mathbf{h}_{\mathrm{base}} \parallel \mathbf{h}_{\mathrm{conf}}^{\mathrm{res}} \parallel \mathbf{h}_{\mathrm{conf}}^{\mathrm{global}} \parallel \phi_s(\mathbf{s}_{\mathrm{conf}})\right],$$

whose final linear layer is initialized to zero. Using the availability indicator c_{conf} defined in Eq. (10), the effective residual and the final structural-expert logit are

$$\begin{aligned} \lambda_{\mathrm{conf}} &= \sigma\left(\beta_{\mathrm{conf}}\right), \\ \Delta z_{\mathrm{conf}} &= c_{\mathrm{conf}}\, \lambda_{\mathrm{conf}}\, \Delta z_{\mathrm{conf}}^{\mathrm{raw}}, \\ z_{\mathrm{SE3D}} &= z_{\mathrm{base}} + \Delta z_{\mathrm{conf}}, \qquad c_{\mathrm{conf}} = 0 \;\Rightarrow\; z_{\mathrm{SE3D}} = z_{\mathrm{base}}. \end{aligned} \tag{21}$$

where the effective conformer weight is initialized to 0.25 at the beginning of training through $\beta_{\mathrm{conf}}^{(0)} = \mathrm{logit}(0.25)$, so that $\sigma(\beta_{\mathrm{conf}}^{(0)}) = 0.25$. Zero-filled feature tensors alone would not guarantee a zero correction because the residual network may contain nonzero biases; the additional hard availability mask ensures an exact fallback to the single-structure prediction. The conformer branch is permitted only to refine the single-structure predictor rather than replace the geometric backbone, reducing the risk of overfitting to limited and uneven ensemble coverage.

2.4.6 pLDDT-Aware Sequence–Structure Cross-Attention

After the EGNN and the chemical-site residual are applied, the model obtains the chemical-enhanced spatial representation

$$\mathbf{H}^{\mathrm{sp}} = [\mathbf{h}_1^{\mathrm{sp}}, \ldots, \mathbf{h}_L^{\mathrm{sp}}]^\top \in \mathbb{R}^{L \times 128},$$

where $\mathbf{h}_i^{\mathrm{sp}}$ is defined in Eq. (19). Spatial information alone does not determine whether a structural pattern is relevant to the current sequence context. For example, similar hydrophobic contacts may have different implications depending on residue identities and sequence positions. GHXTox therefore uses pLDDT-aware cross-attention within the structural expert, allowing contextualized sequence states to selectively retrieve spatial evidence.

A node-level gate is first constructed for each residue from the 42-dimensional residue descriptors and the pLDDT score:

$$\begin{aligned} g_i^{\mathrm{base}} &= \phi_{\mathrm{node}}\left[\mathbf{f}_i^{\mathrm{res}} \parallel q_i\right], \\ \mathbf{c}_i^{\mathrm{node}} &= \left[\frac{q_i - \mu_q}{T_q},\ \frac{(q_i - \mu_q)^2}{T_q^2},\ \mathbb{I}(q_i < 0.55)\right], \\ r_i^{\mathrm{node}} &= 0.5 + \sigma\left(\mathbf{w}_n^\top \mathbf{c}_i^{\mathrm{node}} + b_n\right), \\ g_i^{\mathrm{node}} &= \mathrm{clamp}\left(g_i^{\mathrm{base}}\, r_i^{\mathrm{node}},\ 0, 1\right), \end{aligned} \tag{22}$$

where $\mu_q = 0.70$, $T_q = 0.12$, and the residual confidence vector $\mathbf{c}_i^{\text{node}}$ increases the sensitivity of the gate to the distribution of structural confidence. The spatial representation is then continuously down-weighted as $\widehat{\mathbf{H}}^{\text{sp}} = \mathbf{H}^{\text{sp}} \odot \mathbf{g}^{\text{node}}$, with the node-level gate vector $\mathbf{g}^{\text{node}} \in [0,1]^L$ broadcast across the feature dimension.

In the cross-attention operation, the contextualized sequence representation serves as the query, and the gated spatial representation serves as the key and value:

$$\mathbf{H}^{\text{cross}} = \text{MHA}\left(Q = \mathbf{H}^{\text{seq}}, K = \widehat{\mathbf{H}}^{\text{sp}}, V = \widehat{\mathbf{H}}^{\text{sp}}\right). \tag{23}$$

This direction has a direct interpretation: each residue position retrieves spatial evidence according to its contextual sequence representation. The operation is not a direct feature interaction between the LoRA-ProtT5–Mamba expert and the structural expert; it occurs internally between the ESM2-enhanced sequence states and the geometric states of the structural expert.

In addition to residue-level gating, a peptide-level gate is generated from the overall structure-confidence profile. The model first summarizes the profile by the mean pLDDT $\bar{q}$, the minimum pLDDT $q_{\min}$, and the low-confidence fraction $r_{\text{low}} = \frac{1}{L}\sum_{i=1}^{L} \mathbb{I}(q_i < 0.55)$, and constructs the peptide-level confidence vector $\mathbf{c}^{\text{pep}} = [(\bar{q} - \mu_q)/T_q,\ (q_{\min} - \mu_q)/T_q,\ r_{\text{low}}]$. The base peptide-level gate and its residual-adjusted form are

$$\begin{aligned} g_{\text{pep}}^{\text{base}} &= \phi_{\text{global}}\left[\bar{q},\ q_{\min},\ r_{\text{low}}\right], \\ r_{\text{pep}} &= 0.5 + \sigma\left(\mathbf{w}_{\text{pep}}^{\top}\mathbf{c}^{\text{pep}} + b_{\text{pep}}\right), \\ g_{\text{pep}} &= \text{clamp}\left(g_{\text{pep}}^{\text{base}}\, r_{\text{pep}},\ 0, 1\right), \end{aligned} \tag{24}$$

where the peptide-level residual scale is analogous to, but separately parameterized from, the node-level residual scale, and $g_{\text{pep}} \in [0,1]$ denotes the scalar peptide-level structural gate.

The final residue-level fusion representation combines the sequence state with the gated structural context, $\mathbf{H}^{\text{fused}} = \text{LN}\left[\mathbf{H}^{\text{seq}} \parallel g_{\text{pep}}\, \mathbf{H}^{\text{cross}}\right]$. The two gating levels answer complementary questions: how reliable the structural evidence is at an individual residue position, and how strongly the structural branch should be trusted for the peptide as a whole. The sequence representation is always kept, whereas the structural context is dynamically scaled according to prediction confidence. Consequently, when the predicted structure has low overall confidence, the structural expert can still rely on its internal sequence-semantic pathway without allowing uncertain coordinates to dominate the classification.

2.4.7 Global Feature Integration and Structural Classification

After pLDDT-aware cross-attention, each valid residue is represented by a 256-dimensional fused vector obtained by concatenating the 128-dimensional internal sequence state with the 128-dimensional structural context. Masked mean pooling is applied over the valid residue positions, and the pooled vector is concatenated with five global physicochemical descriptors before a three-layer MLP produces the base structural logit:

$$\begin{aligned} \mathbf{h}_{\text{pool}} &= \frac{\sum_{i=1}^{L} m_i\, \mathbf{H}_i^{\text{fused}}}{\sum_{i=1}^{L} m_i} \in \mathbb{R}^{256}, \\ \mathbf{h}_{\text{cls}} &= \left[\mathbf{h}_{\text{pool}} \parallel \mathbf{f}_{\text{pep}}^{\text{phys}}\right] \in \mathbb{R}^{261}, \\ z_{\text{base}} &= \phi_{\text{cls}}\left(\mathbf{h}_{\text{cls}}\right), \end{aligned} \tag{25}$$

where the mask excludes padding positions from the pooled representation, and $\mathbf{f}_{\text{pep}}^{\text{phys}} \in \mathbb{R}^5$ contains the normalized peptide length, the length-normalized net charge, the aromatic-residue fraction, the cysteine fraction, and the scaled mean hydropathy. ϕ_{cls} is a three-layer MLP with dimensions $261 \to 128 \to 64 \to 1$, each hidden layer applying GELU activation and dropout.

Unlike maximum pooling, masked mean pooling integrates structural evidence distributed across residue positions, which is appropriate when toxicity-related spatial patterns are not confined to a single local region.

Applying the availability-masked conformer correction defined in Eq. (21) yields the final structural-expert logit, and the structural-expert probability is

$$z_{\mathrm{SE3D}} = z_{\mathrm{base}} + c_{\mathrm{conf}}\,\lambda_{\mathrm{conf}}\,\Delta z_{\mathrm{conf}}^{\mathrm{raw}}, \qquad p_{\mathrm{SE3D}} = \sigma\left(z_{\mathrm{SE3D}}\right),$$

as defined in Eq. (21). No standalone structural threshold is used to generate the final GHXTox class at this stage. Instead, p_{SE3D} is passed directly to the top-level dual-expert fusion described in Section 2.2.3, where it is combined with the calibrated 1D expert probability p_{1D} using frozen fusion weights. A standalone structural threshold is used only for structural-branch ablations or independent expert evaluation and does not determine the final GHXTox prediction.

The structural expert can therefore be summarized as

$$\begin{aligned}&\text{sequence-enhanced residue semantics} + \text{confidence-aware geometry}\\ &+ \text{full-atom chemical sites} + \text{global physicochemical descriptors}\\ &+ \text{availability-masked conformer statistics} \longrightarrow p_{\mathrm{SE3D}}.\end{aligned}$$

The central contribution is the explicit organization of structural information by spatial scale and reliability. The geometric graph captures global residue relationships, the chemical-site branch resolves local functional moieties, the conformer residual summarizes sampled structural variation, and pLDDT continuously modulates edge propagation, chemical messages, and sequence–structure fusion. This design allows the structural expert to incorporate complementary 3D evidence while explicitly handling structure-prediction uncertainty.

Overall, the structural expert integrates residue-level sequence semantics, confidence-aware geometry, full-atom chemical-site interactions, peptide-level physicochemical properties, and availability-masked conformer statistics. Rather than treating all predicted structural evidence as equally reliable, GHXTox continuously modulates its contribution using residue-level and peptide-level confidence signals. The following section describes the training, out-of-fold model-selection, calibration, and parameter-freezing protocols used to optimize the complete framework.

2.5 Training and Model-Selection Protocol

GHXTox consists of a task-adaptive 1D sequence expert and a sequence-enhanced 3D structural expert. Because the two experts differ substantially in input representation, parameter scale, and optimization behavior, they are not jointly optimized through a single end-to-end objective. Instead, expert-level representations are learned independently, after which probability calibration and top-level fusion are performed using training out-of-fold (OOF) predictions. This strategy allows each expert to be optimized under modality-appropriate settings while separating representation learning, checkpoint selection, probability calibration, fusion-parameter determination, and benchmark evaluation into clearly defined stages.

The training protocol follows three central principles. First, identical or highly similar peptides must not appear simultaneously in a training partition and the corresponding validation partition. Second, every prediction used for calibration, fusion-weight selection, or threshold determination must be generated by a model that was not trained on the corresponding sample. Third, Test1 and Test2 are not used for checkpoint selection, calibrator fitting, fusion search, or decision-threshold optimization in the current pipeline. Their prior observation during project development is disclosed; both sets are therefore treated as frozen retrospective benchmarks.

2.5.1 Group-Aware Five-Fold OOF Generation

The training set contains $N = 6387$ peptides. To reduce performance inflation caused by highly similar sequences, a fixed five-fold manifest is constructed by clustering the training sequences into homology groups. The fold-level homology groups were constructed through a separate sequence-clustering procedure, distinct from the dataset-level redundancy removal described in Section 2.1. The first operation determines which samples remain in the benchmark dataset, whereas the second determines which of these samples must remain in the same cross-validation fold. The training data $\mathcal{D} = \{(\mathcal{S}_i, y_i, g_i)\}_{i=1}^{N}$, $N = 6387$, consist of peptide sequences $\mathcal{S}_i$, toxicity labels $y_i \in \{0, 1\}$, and sequence-homology groups g_i. The data are partitioned into five disjoint validation folds:

$$\begin{aligned} \mathcal{D} &= \bigcup_{k=1}^{5} \mathcal{D}_k, \qquad\qquad \mathcal{D}_k \cap \mathcal{D}_l = \varnothing, \quad k \neq l, \\ g_i = g_j &\Longrightarrow f(i) = f(j), \end{aligned} \tag{26}$$

where $f(i)$ denotes the fold assigned to sample i. All peptides belonging to the same homology group are therefore placed in the same fold, preventing closely related sequences from crossing the training–validation boundary.

For fold k, the model is trained only on the complement $\mathcal{D}_{-k} = \mathcal{D} \setminus \mathcal{D}_k$ and evaluated on the unseen fold $\mathcal{D}_k$. The OOF probability of sample $i \in \mathcal{D}_k$ is defined as

$$\mathcal{D}_{-k} = \mathcal{D} \setminus \mathcal{D}_k, \qquad \widehat{p}_i^{\,\mathrm{OOF}} = F_{\boldsymbol{\theta}^{(-k)}}\left(\mathcal{S}_i\right), \quad i \in \mathcal{D}_k, \tag{27}$$

where $\boldsymbol{\theta}^{(-k)}$ denotes the parameters learned exclusively from $\mathcal{D}_{-k}$. Each training peptide consequently receives exactly one probability from a model that did not observe that peptide during optimization.

For the 1D expert, full seed-specific OOF predictions are generated for the training set, and the three probability vectors are aligned by sample index and averaged as in Section 2.3.4; the resulting OOF ensemble scores are used to fit the Platt calibrator described there.

The structural expert uses the same fixed group-aware folds. Each fold-specific checkpoint produces probabilities only for its held-out validation fold,

$$\widehat{p}_{i,\mathrm{SE3D}}^{\,\mathrm{OOF}} = G_{\boldsymbol{\phi}^{(-k)}}\left(\mathcal{S}_i, \mathcal{X}_i, \mathcal{A}_i, \mathbf{q}_i, \mathcal{C}_i\right), \qquad i \in \mathcal{D}_k,$$

where $\mathcal{X}_i$, $\mathcal{A}_i$, $\mathbf{q}_i$, and $\mathcal{C}_i$ denote the residue geometry, full-atom chemical information, pLDDT confidence, and available conformer-ensemble statistics, respectively; $\mathcal{C}_i$ may be unavailable and is handled by the conformer-availability mask defined in Section 2.4.1.

Before combining the expert-level OOF predictions, the implementation verifies immutable sample indices (`source_index`), peptide sequences, toxicity labels, and fold assignments. Duplicate sample indices or mismatches in sequence, label, or fold assignment cause the fusion procedure to terminate rather than silently aligning prediction files by row order. OOF predictions are treated as **training-only model-selection assets**. Although OOF predictions are generated by models that did not observe the corresponding validation samples, parameters selected from the complete OOF table remain development-stage choices and must be evaluated on independent test sets.

2.5.2 Loss Functions and Class-Imbalance Handling

The 1D and 3D experts are optimized using separate objectives. This prevents uneven structural coverage, conformer availability, and confidence variation from directly perturbing the language-model optimization, and it avoids the large ProtT5 parameter space from dominating the smaller structure-specific residual modules.

Loss for the 1D expert. The sequence expert is trained with the positive-class-weighted binary cross-entropy defined in Section 2.3.3, with the class counts N_+ and N_- independently calculated from each fold's training partition. The pipeline does not use Borderline-SMOTE or sequence point-mutation augmentation; the weighted loss increases the contribution of the minority toxic class without introducing synthetic near-duplicate peptides.

Base geometric and chemical-site stage. The structural expert is optimized in stages. The full-atom chemical-site branch is first initialized from a previously trained pLDDT-aware geometric model, while the remaining backbone is frozen. This stage uses the same positive-class-weighted binary cross-entropy as Section 2.3.3, applied to the base structural logit, with w_+ recalculated from each fold's training partition.

During training, pLDDT-dependent Gaussian coordinate perturbations are applied to reduce sensitivity to local errors in predicted structures. The perturbation scale and the perturbed coordinate for residue i are

$$\sigma_i = \sigma_0 \left[1 + \kappa \left(1 - q_i\right)\right], \qquad \boldsymbol{\varepsilon}_i \sim \mathcal{N}\left(\mathbf{0}, \sigma_i^2 \mathbf{I}_3\right), \qquad \widetilde{\mathbf{x}}_i = \mathbf{x}_i + \boldsymbol{\varepsilon}_i, \tag{28}$$

with $\sigma_0 = 0.12$ Å and $\kappa = 2.0$. The same residue-wise displacement is applied consistently to the corresponding backbone, functional-group, and chemical-site coordinates, thereby preserving their intra-residue relative geometry. No coordinate perturbation is applied during validation or inference. Low-pLDDT residues therefore receive stronger perturbations than high-confidence residues. This augmentation uses pLDDT to modulate the perturbation magnitude but does not treat pLDDT as an exact estimate of coordinate-error variance.

Conformer-ensemble residual stage. During conformer-residual training, the pLDDT-aware geometric backbone, internal sequence encoder, chemical-site branch, and base classifier are frozen. Only the residue-level conformer encoder, the global conformer encoder, the conformer-residual MLP, and the conformer logit-residual parameters are optimized. This stage is initialized from the trained chemical-site checkpoint. To avoid applying a second class-frequency correction on top of the existing base classifier, the current conformer-residual configuration uses unweighted BCE and disables coordinate perturbation, with a residual-magnitude penalty:

$$\mathcal{L}_{\text{conf}} = \text{BCEWithLogits}\left(z_{\text{SE3D}}, y\right) + \lambda_\Delta \frac{\sum_{i=1}^{B} c_i \left|\Delta z_{\text{conf},i}\right|}{\max\left(\sum_{i=1}^{B} c_i,\, 1\right)}, \qquad \lambda_\Delta = 0.005, \tag{29}$$

where the residual-magnitude term is computed only on samples with available conformer information. It limits large logit corrections and encourages the conformer branch to act as a conservative refinement rather than re-learning the complete toxicity decision boundary.

The final conformer-residual layer is zero-initialized, and its output is additionally controlled by the hard `conformer_available` mask. Zero initialization makes the initial model equivalent to the single-structure expert, whereas the hard mask guarantees a zero residual for samples without conformer evidence. Together, these mechanisms provide a stable incremental extension of the structural model.

2.5.3 Optimization, Checkpoint Selection, and Hyperparameter Settings

Both experts are optimized using AdamW [28], but different learning rates, training durations, and checkpoint-selection criteria are employed. All random-number generators are initialized with fixed seeds. Data are shuffled only during training, whereas validation and inference loaders preserve deterministic ordering. Table 2 summarizes the optimization and model-selection settings of both experts.

1D expert. The LoRA-ProtT5 and bidirectional Mamba expert is trained with a batch size of 16, a learning rate of 3×10^{-4}, a weight decay of 1×10^{-2}, a fixed duration of 15 epochs,

Table 2: Main optimization and model-selection settings of the two GHXTox experts.

Module or stage	Batch size	Learning rate	Weight decay	Epochs	Checkpoint rule
1D LoRA-ProtT5 + Bi-Mamba	16	3×10^{-4}	1×10^{-2}	Fixed 15	No early stopping
3D chemical-site refinement	32	5×10^{-4}	1×10^{-4}	Max. 30	Best validation MCC; patience 8
Conformer residual refinement	32	1×10^{-3}	5×10^{-4}	Max. 50	Minimum validation loss; patience 8
Single-structure chemical-site backbone	32	Stage-specific	Stage-specific	Fixed 8	Median fold-selected epoch
Source-aware full2795 conformer residual	32	1×10^{-3}	5×10^{-4}	Fixed 6	OOF-defined refit protocol

dropout of 0.1, FP16 mixed precision, and ProtT5 gradient checkpointing, using random seeds 42, 123, and 2025. This configuration yields approximately 13.1 million trainable parameters, comprising the LoRA updates and the downstream sequence-classification modules, whereas the pretrained base transformations and embeddings remain frozen. Every fold model and every full-data model is trained for a fixed 15 epochs. Validation-based early stopping and a separate learning-rate scheduler are not used. A fixed training budget ensures consistency across folds and random seeds and avoids differences in training duration caused by noisy fold-level validation metrics. After cross-validation, three models are independently trained on the complete training set using the same configuration. The final sequence probability is obtained by averaging these three full-data models rather than selecting the seed with the highest validation performance.

3D chemical-site stage. The full-atom chemical-site refinement uses a batch size of 32, a learning rate of 5×10^{-4}, a weight decay of 1×10^{-4}, a maximum duration of 30 epochs, early-stopping patience of 8 epochs, base structural dropout of 0.22, and chemical-site dropout of 0.1. This stage is initialized from a pLDDT-aware geometric checkpoint, and only the chemical-site branch is made trainable. Training is terminated when the validation MCC does not improve for eight consecutive epochs, and the checkpoint with the highest validation MCC is kept.

Conformer-residual stage. The conformer-ensemble residual stage uses a batch size of 32, a learning rate of 1×10^{-3}, a weight decay of 5×10^{-4}, a maximum duration of 50 epochs, early-stopping patience of 8 epochs, conformer-residual dropout of 0.25, a conformer hidden dimension of 32, an initial conformer weight of 0.25, and a residual L1 coefficient of 0.005. Validation loss is used instead of training accuracy to select the conformer checkpoint, reducing sensitivity to threshold fluctuations arising from limited conformer coverage. Only conformer-related modules are optimized, whereas the geometric and chemical backbone remains frozen.

During structural-expert training, gradients of all trainable parameters are clipped to a maximum norm of 5.0 ($\|\nabla_\phi \mathcal{L}\|_2 \leq 5.0$), limiting unusually large gradients caused by local residual corrections or coordinate perturbations.

When a full-data structural checkpoint is required, the number of refitting epochs is not selected using either benchmark test set. Instead, a fixed epoch count is obtained from the median of the fold-specific selected epochs. For the single-structure chemical-site backbone, the recorded fold-level epochs are 8, 8, 14, 2, and 4, whose median is 8, and the full-data refit is trained for eight fixed epochs. The source-aware conformer residual uses the full2795 cohort and a six-epoch full-data refit fixed from the training-side protocol. Neither the refit duration nor the checkpoint was selected using Test1 or Test2.

2.5.4 OOF-Based Calibration, Fusion-Weight Selection, and Threshold Freezing

After the expert-level OOF predictions have been generated, model selection proceeds at the probability level. No ProtT5, Mamba, EGNN, chemical-site, or conformer-residual parameters

are updated during this stage. Only the training OOF probabilities are used to determine the calibrator, fusion weight, and final decision threshold.

The three seed-specific 1D OOF probabilities are first averaged and calibrated using the monotonic Platt mapping described in Section 2.3.4. Platt calibration is applied only to the 1D ensemble. The structural-expert probability is not subjected to an additional top-level calibrator, and no new calibrator is fitted after the final fusion.

The calibrated sequence probability and the structural-expert probability are combined in probability space as in Section 2.2.3, where w denotes the 1D expert weight and $1 - w$ the structural-expert weight. No additional stacking classifier or sample-specific routing network is fitted at this stage, reducing the risk of second-level overfitting to the finite OOF table.

For this top-level fusion, $p_{i,\mathrm{SE3D}}^{\mathrm{OOF}}$ is the pure `structure_probability` emitted by the pLDDT-aware structural expert. It is not the probability of an earlier sequence–structure fusion model and contains no contribution from the current or historical 1D predictor. This strict interface prevents duplicated sequence evidence and avoids mixing old and new model generations during OOF weight selection.

For a candidate weight w and threshold τ, the predicted class is $\hat{y}_i(w, \tau) = \mathbb{I}[p_i(w) \geq \tau]$, and the frozen parameters are selected according to

$$(w^*, \tau^*) = \arg\max_{w,\tau} \ \mathrm{MCC}\left(\mathbf{y}, \hat{\mathbf{y}}(w, \tau)\right). \tag{30}$$

MCC is used as the primary objective because of the pronounced class imbalance in the benchmark data, particularly in Test2. Balanced accuracy was used as the secondary selection criterion when candidate operating points produced identical MCC values under the evaluated search procedure.

The frozen values are $w_{\mathrm{1D}}^* = 0.72$, $w_{\mathrm{SE3D}}^* = 0.28$, and $\tau^* = 0.47022256531284556$ (rounded to 0.4702 when reported). The final probability and class prediction follow the fusion and decision rules in Sections 2.2.3 and 2.6.

A second-level leave-one-fold-out analysis is additionally performed as a stability diagnostic. In each iteration, w and τ are selected using four OOF folds and applied to the remaining fold. This analysis assesses whether the fusion parameters are disproportionately dependent on a particular fold. However, because every upstream prediction stream is not fully regenerated within a separately nested outer loop, this diagnostic is not described as strict nested external validation.

The final values of 0.72/0.28 and 0.47022256531284556 are frozen from the complete training OOF table; the corresponding fused OOF metrics describe training-stage model selection rather than benchmark test-set performance. After freezing, the same calibration parameters, fusion weights, and threshold are applied unchanged to Test1 and Test2. Neither Test1 nor Test2 labels are used for recalibration or parameter adjustment. The frozen parameters, input prediction files, fusion space, and decision threshold are stored in machine-readable files to permit verification of the final evaluation protocol.

2.6 Evaluation Metrics

To comprehensively evaluate GHXTox for peptide toxicity prediction, model performance is assessed from three complementary perspectives: classification performance at a fixed operating threshold, ranking ability based on continuous probability scores, and probabilistic calibration. Reliance on a single metric may conceal important differences in minority-class recognition, false-positive control, or probability reliability. We therefore jointly report confusion-matrix metrics, the area under the receiver operating characteristic curve (AUROC), the area under the precision–recall curve (AUPRC), the Brier score, and the expected calibration error (ECE).

Let the evaluation set contain N peptides with labels $y_i \in \{0, 1\}$ and predicted probabilities $p_i = \mathbb{P}(y_i = 1 \mid \mathcal{S}_i)$, where $\mathcal{S}_i$ denotes the peptide sequence. At the fixed decision threshold τ, the

binary prediction is $\hat{y}_i = \mathbb{I}[p_i \geq \tau]$. For the final GHXTox model, the threshold is frozen using the complete training OOF table (Section 2.5.4), with full-precision value $\tau^* = 0.47022256531284556$ (rounded to 0.4702 in result tables) and applied unchanged to Test1 and Test2. Threshold-dependent metrics are calculated from $\hat{y}_i$; AUROC, AUPRC, the Brier score, and ECE are calculated from the continuous probabilities p_i.

Because Test2 contains only 46 toxic peptides and 536 non-toxic peptides, its class distribution is strongly imbalanced. Accuracy alone may therefore provide an incomplete description of toxic-peptide recognition. Balanced accuracy (BACC) and the Matthews correlation coefficient (MCC) are emphasized as balanced summary measures; sensitivity (SN) and specificity (SP) are reported to expose the trade-off between missed toxic peptides and false alarms; and AUPRC and calibration metrics are reported together with AUROC.

2.6.1 Threshold-Dependent Classification Metrics

At a fixed decision threshold, predictions are summarized by four confusion-matrix counts: true positives (TP), true negatives (TN), false positives (FP), and false negatives (FN). We report accuracy ACC = (TP + TN)/N, sensitivity SN = TP/(TP + FN), specificity SP = TN/(TN+FP), precision Precision = TP/(TP+FP), the F1 score F_1 = 2TP/(2TP+FP+FN), balanced accuracy BACC = (SN + SP)/2, and the Matthews correlation coefficient

Matthews correlation coefficient.

$$\mathrm{MCC} = \frac{\mathrm{TP} \cdot \mathrm{TN} - \mathrm{FP} \cdot \mathrm{FN}}{\sqrt{(\mathrm{TP} + \mathrm{FP})(\mathrm{TP} + \mathrm{FN})(\mathrm{TN} + \mathrm{FP})(\mathrm{TN} + \mathrm{FN})}}, \tag{31}$$

with $-1 \leq \mathrm{MCC} \leq 1$; values of 1, 0, and -1 indicate perfect, chance-level, and completely inverted agreement, respectively. Because MCC jointly accounts for all four counts, it was used as the primary fixed-threshold summary metric and as the objective for OOF fusion-parameter selection [29].

ACC is dominated by the majority class in strongly imbalanced datasets and is therefore not used as the sole measure of performance. SN and SP should be interpreted jointly: lowering the threshold generally increases SN at the cost of additional false positives, whereas raising it has the opposite effect. Precision depends on class prevalence, and F1 balances positive-prediction purity with toxic-peptide recall without explicitly including TN. BACC gives equal weight to the two classes, preventing the larger class from dominating the score. Numerical safeguards were applied when metric denominators were zero; all threshold-dependent metrics used the same OOF-frozen decision threshold (Section 2.5.4), without metric- or test-set-specific optimization.

2.6.2 Ranking and Probability-Calibration Metrics

Fixed-threshold metrics describe classification at one operating point but do not fully characterize sample ranking or probability reliability. We therefore additionally report AUROC, AUPRC, the Brier score, and ECE-10. For a varying threshold t, the true-positive rate TPR(t) and false-positive rate FPR(t) define the ROC curve; AUROC is the area under this curve and measures the probability that a randomly selected toxic peptide is ranked above a randomly selected non-toxic peptide. AUROC is independent of the fixed decision threshold and insensitive to class prevalence, so AUPRC is reported as a complementary ranking metric. AUPRC is computed as average precision AP = $\sum_n (R_n - R_{n-1}) P_n$ over score thresholds sorted in descending order, where P_n and R_n are the precision and recall at threshold n; this value is not necessarily identical to a trapezoidal integral of the PR curve. Because the chance-level PR baseline depends on positive-class prevalence, AUPRC values should be interpreted within, rather than across, datasets. AUROC and AUPRC are reported only when both classes are present in the evaluation subset. Recent studies in peptide toxicity prediction commonly report ROC and PR analyses together [2–4, 7].

Brier score. The Brier score is the mean squared difference between the predicted probability and the binary ground-truth label [30]:

$$\text{Brier} = \frac{1}{N}\sum_{i=1}^{N}(p_i - y_i)^2 . \tag{32}$$

It ranges from 0 to 1 and penalizes both incorrect predictions and excessive confidence; because it reflects calibration and discrimination jointly, it is not interpreted as a pure calibration measure.

Ten-bin expected calibration error. To further quantify the agreement between predicted confidence and observed toxicity frequency, we calculate the expected calibration error using ten equal-width probability bins, denoted ECE-10 [31]. Predictions are divided into ten equal-width bins over $[0, 1]$, with the final bin including $p_i = 1$. Let $\mathcal{B}_m$ denote the m-th bin, with mean confidence $\text{conf}(\mathcal{B}_m)$ and observed toxic-peptide frequency $\text{freq}(\mathcal{B}_m)$. ECE-10 is defined as

$$\text{ECE}_{10} = \sum_{m=1}^{10}\frac{|\mathcal{B}_m|}{N}\,\left|\text{conf}\left(\mathcal{B}_m\right) - \text{freq}\left(\mathcal{B}_m\right)\right| . \tag{33}$$

Values closer to zero indicate better agreement between predicted confidence and observed toxicity frequency; for example, a well-calibrated group of predictions near 0.8 should contain approximately 80% toxic peptides. Because ECE depends on the binning scheme, the number and boundaries of the bins remain fixed across models and datasets.

Joint interpretation of the metrics. No individual metric is sufficient to characterize peptide-toxicity prediction; the reported measures are therefore interpreted jointly. ACC summarizes overall correctness; SN and SP quantify toxic-peptide misses and non-toxic false positives, respectively; BACC evaluates class-balanced recognition; MCC provides a comprehensive correlation-based measure using all four confusion-matrix terms; AUROC and AUPRC assess ranking performance; and the Brier score and ECE-10 characterize probabilistic reliability. Exact numerical tables are used as the primary form of model comparison, with ROC and PR curves presented together. Radar charts, when included, are treated only as secondary summaries and do not replace exact metric tables or uncertainty analysis.

2.7 Robustness, Uncertainty, and Statistical Analysis

A single performance point estimate cannot fully characterize the reliability of a deep learning predictor. Model performance may be affected by random initialization, finite evaluation samples, class imbalance, predicted- structure quality, and distribution shift. The GHX-Tox evaluation framework therefore prespecifies five complementary perspectives: multi-seed reproducibility, class-stratified bootstrap confidence intervals, paired significance testing between frozen models, training-OOF conformer information-ceiling diagnostics, and subgroup and structure-quality analysis. For the multi-conformer benchmark output, inferential statistics are reported only after they have been regenerated from the corresponding sample-level prediction vectors; intervals from preceding model variants are not reused.

All analyses in this section are conducted after the model architecture, checkpoints, probability-calibration parameters, fusion weights, and decision threshold have been frozen. No ProtT5, Mamba, EGNN, chemical-site, or conformer-residual parameters are updated during statistical evaluation, and the threshold is not modified according to the resulting confidence intervals or significance tests. This design explicitly separates model development from frozen-model evaluation.

Three sources of uncertainty are distinguished. Multi-seed analysis characterizes variation arising from stochastic optimization; bootstrap intervals quantify finite-sample uncertainty conditional on a frozen predictor; and subgroup analyses examine whether predictive performance is concentrated within particular sequence-length, physicochemical, or structure-confidence ranges.

These analyses provide complementary evidence and should not be interpreted as interchangeable measures of uncertainty.

2.7.1 Multi-Seed Reproducibility and Prediction Dispersion

Neural-network training may vary because of parameter initialization, mini-batch ordering, dropout masks, and mixed-precision computation. To assess the sensitivity of the 1D expert to these factors, three LoRA-ProtT5 and bidirectional Mamba models are trained using the random seeds $\mathcal{R} = \{42, 123, 2025\}$, while keeping the data partitions, architecture, training duration, and all hyperparameters unchanged. The current 1D expert uses an equal-weight probability average of the three full-data models rather than selecting the best-performing seed post hoc. The top-level model uses a single frozen 3D structural expert: the evaluated multi-checkpoint 3D ensemble did not satisfy the prespecified improvement criterion and was therefore not adopted in the final GHXTox model.

For an evaluation metric M, seed-specific results may be summarized as $\bar{M} \pm \mathrm{SD}(M)$, where $\bar{M}$ and $\mathrm{SD}(M)$ denote the mean and sample standard deviation across the three seeds. All seed-specific models must be evaluated using the same frozen threshold, or using thresholds selected without access to the corresponding test set; a test-specific optimal threshold must not be selected separately for each seed.

Prediction dispersion can also be quantified at the sample level. For peptide i, let $p_{i,\mathrm{1D}}^{(s)}$ denote the uncalibrated probability from seed s; the uncalibrated three-seed average $\bar{p}_{i,\mathrm{1D}}^{\mathrm{raw}}$ is distinguished from the training-time OOF average $\bar{p}_{i,\mathrm{1D}}^{\mathrm{OOF}}$ defined in Section 2.5.1, and the seed dispersion $u_{i,\mathrm{seed}}$ is the sample standard deviation of the three seed probabilities. A large $u_{i,\mathrm{seed}}$ indicates that the prediction is sensitive to optimization randomness and may be used to flag boundary cases or expert-disagreement samples. It is, however, an empirical measure of training variability rather than a Bayesian posterior uncertainty estimate or a direct probability of prediction error.

The final 1D pathway follows the order defined in Section 2.3.4:

$$\left\{p_{i,\mathrm{1D}}^{(s)}\right\}_{s=1}^{3} \longrightarrow \bar{p}_{i,\mathrm{1D}}^{\mathrm{raw}} \longrightarrow \text{OOF-fitted Platt calibration} \longrightarrow p_{i,\mathrm{1D}}^{\mathrm{cal}}.$$

Seed-specific metrics characterize training stability, whereas the calibrated three-seed average is used for top-level 1D/3D fusion. The final calibrated ensemble prediction is a deterministic function of the three trained seed models; seed-level variability is therefore summarized by $u_{i,\mathrm{seed}}$ across the constituent models rather than by assigning a standard deviation of zero to their averaged prediction.

2.7.2 Class-Stratified Bootstrap Confidence Intervals

Performance metrics calculated on a finite test set are themselves estimates. This issue is particularly important for Test2, where a small number of changes in TP or FN can substantially alter SN, BACC, and MCC. We therefore use class-stratified percentile bootstrap resampling to calculate 95% confidence intervals [32].

Let $\mathcal{I}_+$ and $\mathcal{I}_-$ denote the positive- and negative-class index sets. Each replicate independently resamples the two classes with replacement, preserving their original sizes, and concatenates the two index vectors into $\mathbf{i}^{(b)}$; the index vectors preserve repeated indices, avoiding extreme class-ratio fluctuations or single-class replicates when the positive class is small. For each replicate, the metric is recomputed on the resampled rows, $M^{(b)} = M\left(\mathbf{y}_{\mathbf{i}^{(b)}},\ \mathbf{p}_{\mathbf{i}^{(b)}}\right)$. The primary test-set analysis used $B = 5000$ bootstrap replicates, whereas subgroup-difference analyses used 1000 replicates. The 95% percentile confidence interval is formed from the empirical 2.5th and 97.5th percentiles of the B replicate values, $[Q_{0.025}, Q_{0.975}]$, where Q_α denotes the empirical α-quantile.

For ACC, BACC, SN, SP, F1, and MCC, each bootstrap replicate uses the original frozen model threshold; the threshold is not reoptimized within any resampled data set. AUROC, AUPRC, and the Brier score are calculated directly from the resampled continuous probabilities.

These intervals quantify finite-sample uncertainty conditional on the frozen predictor and the observed evaluation distribution; they do not incorporate training-seed variability or distribution shift. Bootstrap uncertainty is therefore interpreted separately from the multi-seed analysis in Section 2.7.1 and the subgroup analysis in Section 2.7.5.

2.7.3 Paired Bootstrap and Randomization Significance Testing

When the current GHXTox model is compared with the previous model or another frozen internal baseline, both models produce predictions for the same peptides; the sample-level predictions are therefore paired. Comparing two independent confidence intervals would ignore this dependence. We instead analyze the metric difference directly on aligned samples.

Let M_A and M_B denote the metric values of frozen models A and B, respectively, where B is the current model and A is the frozen baseline; the observed difference is $\Delta_M^{\text{obs}} = M_B - M_A$. For BACC, SN, SP, F1, MCC, AUROC, and AUPRC, a positive difference favors model B; for lower-is-better measures such as the Brier score and ECE, a negative difference indicates improvement.

Paired class-stratified bootstrap. For every bootstrap replicate, the same class-stratified index vector $\mathbf{i}^{(b)}$ defined in Section 2.7.2 is applied to both models: $\Delta_M^{(b)} = M(\mathbf{y}_{\mathbf{i}^{(b)}},\ \mathbf{p}_{B,\mathbf{i}^{(b)}}) - M(\mathbf{y}_{\mathbf{i}^{(b)}},\ \mathbf{p}_{A,\mathbf{i}^{(b)}})$, where the identical indices preserve the paired sample structure. The 95% percentile interval is formed from the empirical 2.5th and 97.5th percentiles of the $\Delta_M^{(b)}$ values. For a higher-is-better metric, an interval entirely above zero supports a consistent positive difference; for a lower-is-better metric such as the Brier score, improvement corresponds to an interval entirely below zero.

Paired randomization test. A within-sample exchange randomization test is additionally used to evaluate the null hypothesis that the two frozen model outputs are exchangeable for each peptide [33]. For randomization replicate r, $z_i^{(r)} \sim \text{Bernoulli}(0.5)$ is independently drawn for each sample; if $z_i^{(r)} = 1$, the probabilities and frozen-threshold decisions of models A and B are exchanged for sample i, and otherwise they remain unchanged. The resulting null difference is $\Delta_{M,0}^{(r)} = M_B^{(r)} - M_A^{(r)}$. A two-sided paired randomization test was performed by comparing the absolute randomized metric differences with the absolute observed difference:

$$p_{\text{rand}} = \frac{1 + \sum_{r=1}^{R} \mathbb{I}\left(\left|\Delta_{M,0}^{(r)}\right| \geq \left|\Delta_M^{\text{obs}}\right|\right)}{R+1}. \tag{34}$$

The primary paired analysis uses $B = 5000$ and $R = 5000$ replicates with a fixed seed of 2026. The implementation exchanges both continuous scores and the corresponding frozen-threshold decisions within each sample.

MCC and BACC are treated as prespecified primary summary measures, whereas SN, SP, AUROC, AUPRC, and the Brier score provide complementary evidence. Because several correlated metrics are examined for the same model comparison, secondary p-values are interpreted descriptively rather than as multiple independent discoveries.

Finally, Test1 and Test2 have been repeatedly observed during the historical development of the project. When recomputed for the current prediction vectors, paired confidence intervals and randomization p-values on these sets provide retrospective evidence for differences between frozen models, rather than fully independent prospective confirmation; this limitation is explicitly recorded by the project validation implementation.

2.7.4 Training-OOF Conformer Information-Ceiling Diagnostics

To distinguish limited conformer diversity from limited use of that diversity by the frozen predictor, we conducted two post hoc diagnostics using only the group-aware training OOF folds. Neither diagnostic accessed Test1 or Test2, altered the full2795 checkpoint, or participated in model selection.

First, each of the eight conformers available for 2,795 covered training peptides was evaluated with the corresponding frozen fold-specific structural expert, yielding 22,360 held-view predictions. For each peptide, we summarized the maximum pairwise conformer RMSD, mean residue RMSF, and range of the eight structural-expert probabilities. Label-free fixed aggregators, including the mean, median, trimmed mean, confidence- and centrality-weighted means, and a structural-medoid rule, were compared with the frozen single-structure output under the unchanged top-level fusion and decision threshold. A label-aware choice among the single-structure and per-conformer outputs was computed only as an upper-bound diagnostic that cannot be used for prediction.

Second, pooled spatial embeddings were extracted for every conformer using the same fold-specific frozen expert. A compact 50,113-parameter permutation-invariant residual head was fitted under fold-local train/held separation, while all upstream encoders, the 1D probability stream, the top-level fusion weights, and the decision threshold remained fixed. The five held-fold prediction vectors were concatenated once for paired OOF evaluation and class-stratified bootstrap analysis. Read-only diagnostics also quantified the Pearson correlation between the 1D and base 3D probabilities, the correlation between the learned residual and the correction required by the observed label, and the fraction of residual directions that moved the fused probability toward that label. These quantities were used to characterize a possible representation–label alignment bottleneck, not to tune the residual head.

2.7.5 Subgroup and Structure-Quality Analysis

Aggregate test metrics may conceal performance heterogeneity across peptide lengths, physicochemical properties, and structure-quality ranges. This issue is especially important for structure-dependent models: an overall gain may be concentrated among high-pLDDT peptides and may not generalize to short peptides, uncertain structures, or samples with incomplete chemical-site coverage. We therefore conduct prespecified subgroup analyses without modifying the frozen model or threshold.

The primary current-model length analysis uses 11–15, 16–30, and 31–50 residues. General structural diagnostics additionally stratify samples by mean pLDDT (<0.55, 0.55–0.70, and ≥ 0.70), net charge (negative, neutral, and positive), mean hydropathy (<0, 0–1, and ≥ 1), cysteine count (0, 1–2, and ≥ 3), and chemical-site coverage (<0.80, 0.80–0.95, and ≥ 0.95). Chemical-site coverage is defined as the fraction of residues containing at least one valid chemical site:

$$r_{\text{site}} = \frac{1}{L} \sum_{i=1}^{L} \mathbb{I}\left(\text{residue } i \text{ contains at least one valid chemical site}\right). \tag{35}$$

The validation implementation also records the minimum pLDDT, the low-pLDDT residue fraction, the net charge, the mean hydropathy, and the cysteine count.

To reduce unstable conclusions from extremely small subsets, a stratum is reported only when $N_{\text{stratum}} \geq 20$ and $N_{+,\text{stratum}} \geq 5$ and $N_{-,\text{stratum}} \geq 5$. Strata that failed these reporting criteria were recorded with their sample counts and the reason for omission, but were excluded from metric-difference interpretation. These requirements match the current subgroup-validation implementation.

For each eligible stratum, BACC, SN, SP, F1, and MCC are calculated using the same threshold frozen on the full training OOF predictions; AUROC, AUPRC, and the Brier score are

computed from the original continuous probabilities. No stratum-specific threshold optimization is performed.

When comparing the current model with a frozen baseline, the subgroup difference is $\Delta M_g = M_{B,g} - M_{A,g}$, where g denotes a stratum. Class-stratified bootstrap intervals are calculated for the MCC and AUPRC differences, using 1000 subgroup resamples in the current validation protocol.

Subgroup results are treated as heterogeneity diagnostics rather than new model-selection evidence. Because multiple variables and bins are examined, isolated positive or negative findings are interpreted together with the sample size, the positive-class count, the neighboring-stratum trends, and the confidence intervals.

In particular, pLDDT is a confidence estimate produced by a structure-prediction model. Better performance in a high-pLDDT subgroup does not establish correspondence to experimentally correct structures, and poorer performance at low pLDDT does not imply that the corresponding residues lack biological relevance. GHXTox uses continuous confidence weighting rather than excluding low-confidence samples; subgroup analysis is used to characterize applicability boundaries rather than to justify hard filtering.

2.8 Interpretability Analysis Protocol

Interpretability analyses are defined on the frozen model and do not participate in feature selection, threshold selection, or fusion-weight optimization. Unless otherwise stated, probability changes are computed with all non-target inputs held fixed, summarized separately for toxic and non-toxic peptides, and accompanied by peptide-level bootstrap intervals. The analyses therefore describe model sensitivity and expert complementarity; they are not interpreted as evidence of a causal toxicophore.

2.8.1 Fixed-Representation Residue Occlusion and Seed Stability

Residue-level sensitivity was evaluated for a prespecified 16-peptide cohort selected independently of the attribution values. Within each test set and each expert-outcome group (both experts correct, only the 1D expert correct, only the 3D expert correct, or both experts incorrect), one toxic and one non-toxic peptide nearest the frozen fusion threshold were selected. Ties were resolved by source index. This rule prevents favorable cases from being chosen after inspecting the attribution maps.

For each of the three frozen 1D checkpoints, task-adapted ProtT5 hidden states were computed in the original 16-record inference batches, thereby preserving the inference-time padding configuration and the boundary context of the backward convolution. One residue-aligned, post-contextual ProtT5 vector was then set to zero while all other contextual vectors, the sequence length, bidirectional Mamba layers, pooling operation, and classifier parameters were held fixed. The per-seed occlusion effect was defined as the unoccluded logit minus the occluded logit. For every perturbed state, the three raw seed probabilities were averaged before applying the frozen OOF-fitted Platt transform. The displayed signed attribution was the calibrated probability change normalized by the within-peptide L_1 norm. Stability was summarized by residue-wise sign agreement and the pairwise Spearman correlation of absolute logit effects across seeds. This fixed-representation experiment quantifies the sensitivity of the downstream 1D layers to contextual residue vectors; it is neither a token-mutation experiment nor an Integrated Gradients analysis, and it does not test residue-level biological causality.

2.8.2 Chemical-Site Deletion Analysis

For the same prespecified cohort, each valid modeled chemical site was ablated by setting its availability mask, type vector, coordinate, orientation, and orientation-validity entry to zero. All

non-target sequence features, residue geometry, pLDDT values, other chemical sites, conformer descriptors, model weights, and operating thresholds were held fixed. The complete frozen 3D forward pass was recomputed, including the conformer residual when a cache entry was available. Complementary perturbations deleted all sites carrying one chemical category or all site slots assigned to one residue. We report $\Delta p_{3D} = p_{3D}^{\text{base}} - p_{3D}^{\text{deleted}}$ and, with the 1D probability fixed, the induced fusion change $\Delta p_{\text{fused}} = 0.28\Delta p_{3D}$. Because the modeled sites and candidate spatial relations are derived from predicted structures, these quantities describe frozen-model sensitivity rather than confirmed chemical bonds or intervention effects.

2.8.3 pLDDT–Gate Association Analysis

The unperturbed structural expert was additionally run on all 1,126 Test1 and 582 Test2 peptides. For each peptide, we recorded the mean pLDDT, learned global gate, and the within-peptide Spearman correlation between residue-level pLDDT and node gates. Associations were summarized separately by test set and for the pooled descriptive sample. Ninety-five percent bootstrap percentile intervals were obtained from 2,000 peptide-level resamples with seed 2026. This analysis characterizes the association between predicted-structure confidence and learned gate values; it does not establish structural accuracy, feature attribution, or biological causality.

2.8.4 Expert Agreement and Conflict Analysis

Each benchmark peptide is assigned to one of four predeclared groups: both experts correct, only the 1D expert correct, only the 3D expert correct, or both experts incorrect. We compare the sequence probability, structural probability, fused probability, peptide length, class label, and structural confidence across these groups. Expert outcomes use the training-derived standalone thresholds of 0.392 for the 1D expert and 0.677819 for the 3D expert; the fused decision uses the separately frozen threshold of 0.47022256531284556. Cases rescued by structural evidence are inspected alongside cases in which structural evidence reduces an otherwise correct 1D score. This analysis is performed after freezing all model parameters and is used to describe complementarity rather than to retune the fusion rule on Test1 or Test2.

3 RESULTS AND DISCUSSION

3.1 Overall Predictive Performance on the Benchmark Test Sets

Using the OOF-frozen weight and threshold without dataset-specific retuning, GHXTox correctly classified 1,068 of 1,126 Test1 peptides (TP/TN/FP/FN = 287/781/25/33). This corresponds to ACC 0.9485, BACC 0.9329, SN 0.8969, SP 0.9690, F1 0.9082, MCC 0.8726, AUROC 0.9811, and AUPRC 0.9675. On the strongly imbalanced Test2 set, GHXTox correctly classified 552 of 582 peptides (35/517/19/11), yielding ACC 0.9485, BACC 0.8627, SN 0.7609, SP 0.9646, F1 0.7000, MCC 0.6746, AUROC 0.9433, and AUPRC 0.8043. The lower Test2 BACC and F1, despite high ACC, reflect the difficulty of recovering the 46 toxic peptides and demonstrate why class-balanced and precision–recall metrics are essential for this benchmark.

3.1.1 Comparison with Existing Peptide-Toxicity Predictors

Table 3 compares models reported on the same benchmark partitions. Values for ToxMSRC and ToxPLTC are transcribed from the corresponding publications or their accompanying result tables [2, 3]; NR denotes an unreported metric. GHXTox obtains the highest Test1 ACC, BACC, SP, MCC, AUROC, and AUPRC among the compared rows, while ToxPLTC has higher Test1 SN. On Test2, GHXTox obtains the highest ACC, SP, MCC, and reported AUPRC, but its BACC and SN remain below both reference models. The results therefore point to a stronger overall balance of specificity, correlation, and positive-class ranking, not comprehensive superiority on every metric. Because Test1 and Test2 had been observed historically, these differences remain retrospective benchmark evidence and require confirmation on a new prefrozen external set.

Table 3: Performance comparison on the benchmark Test1 and Test2 sets.

Dataset	Model	ACC	BACC	SN	SP	MCC	AUROC	AUPRC
Test1	ToxMSRC	0.9405	0.9217	0.8781	0.9653	0.8520	0.9650	NR
	ToxPLTC	0.9378	0.9302	**0.9125**	0.9479	0.8496	0.9709	0.9426
	GHXTox (current)	**0.9485**	**0.9329**	0.8969	**0.9690**	**0.8726**	**0.9811**	**0.9675**
Test2	ToxMSRC	0.9416	0.8689	0.7826	0.9552	0.6550	0.9430	NR
	ToxPLTC	0.9261	**0.8804**	**0.8261**	0.9347	0.6197	0.9295	0.7353
	GHXTox (current)	**0.9485**	0.8627	0.7609	**0.9646**	**0.6746**	**0.9433**	**0.8043**

3.1.2 Discrimination, Calibration, and Decision Characteristics

The final fused probabilities have Brier scores of 0.0407 and 0.0430 on Test1 and Test2, respectively. These values summarize mean squared probability error under the frozen inference protocol. ECE-10 is not reported here because a consistently regenerated value is not available for both current multi-conformer prediction files; keeping ECE values from the preceding zero-conformer artifact would mix non-identical model outputs. Test2 AUPRC (0.8043) directly reflects the precision–recall trade-off for the rare toxic class, which AUROC does not capture. The 11 false negatives on Test2 also delineate the limits of the frozen operating point; the model's high specificity should not be interpreted as complete minority-class coverage.

Figure 3: Overall discrimination and calibration performance of GHXTox. (A, B) Test1 ROC and PR curves; (C, D) Test2 ROC and PR curves; (E, F) Test1 and Test2 reliability diagrams.

3.2 Complementarity of the 1D and 3D Experts

The two experts are fused only after independent probability generation. The 1D branch provides the stronger standalone ranking signal, whereas the 3D branch contributes confidence-weighted geometry and local chemical context. This separation makes the gain attributable to complementary probability evidence rather than duplicated use of an earlier fused prediction.

3.2.1 Performance of the Individual Experts

Table 4 reports threshold-free discrimination and probability error for probability streams that are aligned with the current current configuration. Platt scaling preserves the 1D ranking but reduces its Brier score. Relative to calibrated 1D alone, final fusion improves Test1 AUROC/AUPRC from 0.9799/0.9644 to 0.9811/0.9675 and reduces Brier from 0.0429 to 0.0407. On Test2, the corresponding values change from 0.9409/0.7831/0.0451 to 0.9433/0.8043/0.0430. The current fused predictor therefore adds complementary structural ranking information while maintaining a lower probability error than the calibrated 1D stream. Historical single-structure SE3D and equal-weight rows are not relabeled as current component results because they were generated before activation of the benchmark conformer cache.

Table 4: Performance of the 1D probability streams and the current probability-level fusion. The final row for each benchmark uses the multi-conformer structural stream.

Dataset	Probability stream	AUROC	AUPRC	Brier
Test1	1D, uncalibrated	0.9799	0.9644	0.0445
	1D, OOF Platt calibrated	0.9799	0.9644	0.0429
	Current fusion (0.72/0.28)	**0.9811**	**0.9675**	**0.0407**
Test2	1D, uncalibrated	0.9409	0.7831	0.0491
	1D, OOF Platt calibrated	0.9409	0.7831	0.0451
	Current fusion (0.72/0.28)	**0.9433**	**0.8043**	**0.0430**

3.2.2 OOF-Frozen Probability Fusion

The 0.72 sequence weight is an empirical result of the training-OOF search, not a manually imposed assertion that sequence information is intrinsically more important. Both expert weights and the full-precision decision threshold of 0.47022256531284556 were fixed before Test1 or Test2 evaluation. The held-out benchmark results therefore assess one prespecified operating point; no benchmark-specific reweighting, threshold adjustment, or post hoc selection between equal and unequal fusion is performed.

3.2.3 Expert Agreement, Disagreement, and Error Correction

The final weighting allows structural evidence to alter borderline rankings while preventing the weaker standalone 3D branch from dominating the decision. The observed corrections are sample-specific and not guaranteed to generalize; inferential uncertainty is not asserted until paired resampling is regenerated from the current prediction vectors.

Figure 4: Complementarity and decision changes induced by expert fusion. (A) scatter of 1D vs SE3D probabilities; (B) error-overlap diagram; (C) probability shifts; (D) corrected/regressed/unchanged counts; (E, F) representative corrected cases.

3.3 Analysis of the Task-Adaptive 1D Sequence Expert

To investigate the role of the task-adaptive 1D expert, we first compared the current sequence architecture with the earlier sequence pipeline based on pooled ProtT5 representations and a TextCNN classifier [3], and further examined task adaptation, bidirectional state-space modeling, multi-seed ensembling, and OOF-based probability calibration. Unlike approaches that use a pretrained protein language model merely as a fixed feature extractor, GHXTox preserves residue-level ProtT5 representations, adapts them to peptide toxicity prediction through LoRA, and subsequently models residue dependencies in both the N-to-C and C-to-N directions using bidirectional Mamba.

The gain of the new 1D expert did not appear uniformly across threshold-dependent metrics. At their respective OOF-frozen operating points, the previous ProtT5-TextCNN sequence model achieved BACC, MCC, AUROC, and AUPRC values of 0.9317, 0.8586, 0.9724, and 0.9436, respectively, whereas the LoRA-ProtT5–Bi-Mamba expert achieved 0.9300, 0.8562, 0.9761, and 0.9533. Thus, BACC and MCC remained broadly comparable, while AUROC and AUPRC increased by approximately 0.0037 and 0.0097, respectively.

The improvement was concentrated in probability ranking. This stronger ranking stream benefits subsequent fusion with structural evidence and explains why sequence-model comparisons should not rely on ACC or BACC alone.

3.3.1 Task Adaptation of ProtT5 and the Final 1D Representation

Existing peptide-toxicity predictors commonly employ protein language models to generate fixed sequence representations. For example, StrucToxNet uses ProtT5-derived sequence features within its graph representation, whereas PeptiTox integrates ESM2 embeddings with geometry derived from predicted structures [6, 7]. A major distinction of GHXTox is that ProtT5 is adapted to the toxicity task through LoRA, while its residue-level representations are preserved for downstream sequence modeling.

This design serves two complementary purposes. First, low-rank updates allow the pretrained semantic space to adapt to peptide toxicity prediction without requiring full fine-tuning of the ProtT5 backbone. Second, peptide-level pooling is deliberately postponed: contextualized residue states are passed to the Bi-Mamba encoder, preserving position-dependent information and allowing additional modeling of local and long-range residue dependencies.

Because LoRA and Bi-Mamba were not ablated independently, their contribution is evaluated jointly.

3.3.2 Bidirectional State-Space Modeling of Peptide Sequences

After task-adaptive ProtT5 encoding, GHXTox employs bidirectional Mamba rather than TextCNN or immediate peptide-level pooling for secondary sequence modeling. This choice is motivated by the properties of peptides: toxicity-related evidence may not be confined to a single local motif; instead, combinations of charged, hydrophobic, aromatic, or other residues may interact over distant sequence positions. Fixed-size convolutions can capture such dependencies only through progressively enlarged receptive fields, whereas a state-space model propagates contextual information through recurrent state updates.

Although peptide sequences possess a natural N-to-C direction, toxicity classification is not an autoregressive prediction task. GHXTox therefore performs both forward and backward selective scans and integrates their representations. Each residue can consequently be contextualized by information from both termini rather than forcing the classifier to adopt a single direction of information propagation.

Because no strict single-variable comparison between unidirectional and bidirectional Mamba under otherwise identical conditions is available, we do not assign a quantitative

performance gain specifically to bidirectionality. The complete LoRA-ProtT5–Bi-Mamba configuration is therefore evaluated as a whole through group-aware OOF evaluation and controlled comparisons.

3.3.3 Multi-Seed Ensembling, Calibration, and Controlled Negative Results

After the basic sequence architecture had been established, we further investigated model variance and probability reliability. The final 1D expert averages the probabilities of three independently trained full-data models with equal weights; no seed was selected on test performance. The 15-member fold-checkpoint ensemble did not improve the OOF validation criteria and was therefore not used.

Following seed averaging, training OOF predictions were used to fit the monotonic Platt calibrator. This operation primarily adjusts the probability scale without altering the ranking of samples. The observed results were consistent with this role: the 1D Brier score decreased from 0.0445 to 0.0429 on Test1 and from 0.0491 to 0.0451 on Test2. After subsequent fusion with the 3D expert, the Brier scores were further reduced to 0.0407 and 0.0430, respectively. Calibration therefore plays a substantive role in the pipeline: it aligns the probability scales of the two experts before fusion.

Three intuitively attractive extensions were additionally evaluated but not adopted. Conservative positive-class sequence perturbation shifted predictions toward the positive class and degraded specificity and the Brier score. Length-specific calibration or thresholding was unstable because of increased sampling variance within smaller strata and did not consistently outperform the global operating point. Finally, the 15-member fold ensemble did not outperform the three full-data models.

These variants did not improve the final 1D configuration. The three-seed full-data average with OOF calibration represents a simpler and more stable solution and reduces the risk of selective reporting or continued optimization against historically observed test sets.

Table 5: Selected controlled comparisons for the task-adaptive 1D sequence expert. OOF architecture rows and benchmark calibration rows are reported separately and should not be compared across evaluation splits.

Evaluation	Variant	BACC	MCC	AUROC	AUPRC	Brier
Training OOF	Pooled ProtT5–TextCNN	0.9317	0.8586	0.9724	0.9436	NR
Training OOF	LoRA-ProtT5–Bi-Mamba	0.9300	0.8562	**0.9761**	**0.9533**	NR
Test1	Three-seed 1D, uncalibrated	–	–	0.9799	0.9644	0.0445
Test1	Three-seed 1D, OOF Platt	–	–	0.9799	0.9644	**0.0429**
Test2	Three-seed 1D, uncalibrated	–	–	0.9409	0.7831	0.0491
Test2	Three-seed 1D, OOF Platt	–	–	0.9409	0.7831	**0.0451**

3.4 Analysis of the pLDDT-Aware Multi-Scale 3D Structural Expert

Compared with the 1D expert, the principal methodological innovations of GHXTox lie in its structural expert. Previous studies have already demonstrated the value of predicted structures for toxicity prediction. PeptiTox combines ESM2 semantics with geometric descriptors extracted from predicted structures, whereas StrucToxNet further employs an equivariant graph neural network to preserve consistency under rigid transformations of the three-dimensional coordinates [6, 7]. For short peptides, however, the more fundamental question is which structural evidence should be trusted and at which spatial scale it should be used.

The GHXTox structural expert is organized around this question. ESMFold predictions are treated as uncertain structures whose pLDDT scores act as a continuous reliability variable

throughout geometric message passing, residue-level gating, peptide-level gating, and training-time coordinate perturbation. The model then extends beyond Cα-level residue geometry by extracting local chemical pseudo-sites from full-atom PDB structures and, when conformer ensembles are available, incorporating statistics such as RMSF, distance variation, and contact occupancy through a conservative residual pathway.

The 3D contribution therefore lies in a hierarchical structural representation that organizes evidence by reliability and spatial scale, from residue geometry and local side-chain chemistry to sampled conformational variation.

3.4.1 Effect of Structural-Confidence Modeling

Structure-quality analysis first confirmed that pLDDT contains useful information: peptides with higher mean pLDDT generally exhibit stronger predictive performance. However, this does not imply that discarding low-confidence samples is an optimal modeling strategy. In a controlled high-pLDDT-only experiment, restricting training to high-confidence structures substantially reduced performance over the complete test domain, with MCC values of only 0.7232 on Test1 and 0.5593 on Test2.

Hard filtering removed difficult but informative samples along with uncertain structures.

GHXTox therefore uses a continuous confidence-aware strategy. Low-confidence edges remain in the graph but contribute weaker messages; residue-level gates modulate structural evidence at individual positions; peptide-level gating controls the overall use of structural context; and stronger coordinate perturbations are applied to low-pLDDT residues during training.

This differs from approaches that analyze structure quality only after model inference. In GHXTox, pLDDT is an active component of representation learning, not a diagnostic variable appended after inference.

3.4.2 Contribution of Full-Atom Chemical Sites

A Cα-level residue graph effectively describes peptide compactness, nonlocal contacts, and residue neighborhoods but compresses chemically complex side chains into single residue nodes. To determine whether local spatial chemistry provides additional discriminative information, we introduced a full-atom-PDB-derived chemical-site residual on top of the frozen Cα-EGNN. A strict paired five-fold design was adopted: in each fold, the new model was initialized from the same control checkpoint, while the sequence branch, Cα-EGNN, fusion module, and classifier were frozen, leaving only the newly introduced chemical-site branch trainable.

Across the five CD-HIT homology-grouped folds, the complete chemical-site model increased mean BACC from 0.9153 to 0.9171 ($\Delta = +0.0018$), F1 from 0.8790 to 0.8814 ($\Delta = +0.0023$), MCC from 0.8312 to 0.8342 ($\Delta = +0.0030$), AUROC from 0.9687 to approximately 0.9692, and AUPRC from 0.9388 to 0.9401 ($\Delta = +0.0013$). More importantly, MCC, F1, and AUPRC improved in all five folds.

Although modest in magnitude, the improvements were consistent across folds. Because only the chemical-site branch was trainable, the result indicates that local side-chain chemistry provides a small but reproducible signal beyond residue-centered geometry.

Component-level ablations further support this interpretation. On the fixed fold-0 subset, removing the second chemical site, orientation invariants, either distance scale, explicit compatibility channels, pLDDT weighting, or zero initialization decreased MCC. Keeping only the raw-distance or only the normalized-distance representation reduced MCC by approximately 0.0059 and 0.0058, respectively, whereas removing interaction-type channels or pLDDT weighting each reduced MCC by approximately 0.0057.

PeptiTox demonstrated the utility of geometric information from predicted structures, whereas ToxiPep highlighted complementarity between local atomic chemistry and sequence context [4, 6]. GHXTox integrates these ideas within a confidence-aware 3D structural expert. The

resulting interaction channels should nevertheless be interpreted as candidate spatial-chemical relationships available to the model, rather than confirmed hydrogen bonds, salt bridges, cation–π interactions, or disulfide bonds.

3.4.3 Sequence–Structure Interaction and Confidence Gating

Local geometric evidence becomes meaningful only in the appropriate sequence context. The same spatial distance or hydrophobic contact may have different implications depending on residue identity and sequence position. GHXTox therefore avoids simply concatenating sequence and structural vectors after peptide-level pooling. Instead, contextualized sequence states act as queries that selectively retrieve evidence from pLDDT-gated spatial residue representations.

This idea is conceptually related to the context–chemistry alignment emphasized by ToxiPep, but in GHXTox the interaction occurs within a predicted three-dimensional structural expert and is further regulated by both residue-level and peptide-level structural-confidence gates.

The residue-level gate modulates how strongly the structural representation of each residue contributes to the subsequent interaction, whereas the peptide-level gate controls the overall contribution of structural context. Structural information is therefore not switched off when confidence decreases; rather, its contribution is continuously attenuated at multiple levels.

Together with the negative result of hard high-pLDDT-only training described in Section 3.4.1, these observations suggest that the value of this design is the learnable compromise it establishes between preserving structural information and attenuating uncertain evidence.

Because no complete single-variable ablation of cross-attention and each confidence gate under otherwise identical conditions is available, the overall structural gain is not quantitatively decomposed among these components. The evidence therefore supports the reliability-aware interaction mechanism, comprising continuous pLDDT weighting, sequence–structure cross-attention, and hierarchical gating.

3.4.4 Effect of Conformer Statistics

A single ESMFold prediction provides only one deterministic structural anchor, whereas short peptides may exhibit substantial conformational heterogeneity. GHXTox does not treat every generated conformer as an additional labeled training example, nor does it repeatedly execute the complete EGNN for every conformation. Instead, conformer ensembles are compressed into statistics including RMSF, pairwise-distance variation, contact occupancy at 6, 8, and 10 Å, conformer RMSD, and radius of gyration, and these statistics are allowed to produce only a zero-initialized logit residual on top of the single-structure predictor.

The source-aware branch covers 2,795 of the 6,387 training peptides. At the unchanged 0.72/0.28 fusion and decision threshold, activating the source-aware residual increased training-OOF MCC from 0.87255 to 0.87445 and BACC from 0.93220 to 0.93292, while reducing the Brier score from 0.04316 to 0.04309. The MCC difference was +0.00190 under 5,000 class-stratified paired bootstrap resamples, with a 95% interval of [−0.00043, 0.00454] and a paired-randomization p value of 0.203. AUROC and AUPRC decreased by 0.00014 and 0.00015, respectively.

The gain remained limited: conformers showed substantial geometric variation, whereas frozen-expert probabilities changed only marginally. This suggests that the current bottleneck lies more in mapping conformational variation to toxicity labels than in generating geometrically distinct conformers. For peptides without valid conformer statistics, $\Delta z_{\mathrm{conf}} = 0$, and the prediction falls back exactly to the single-structure pathway (alternative aggregators, spatial-set residuals, and training-only optimization variants are detailed in the Supporting Information).

3.4.5 Overall Performance of the Structural Expert

After combining all components, the 3D-v2 structural expert achieved BACC, MCC, AUROC, and AUPRC values of 0.9208, 0.8464, 0.9711, and 0.9474 on Test1, compared with 0.9179, 0.8458, 0.9699, and 0.9426 for the historical Cα-only 3D-v1 model. Test1 therefore showed a consistent but modest improvement. Test2 exhibited a different trade-off: AUPRC increased from 0.6969 to 0.7241, whereas MCC decreased from 0.6320 to 0.6128.

This trade-off indicates that finer-grained structural modeling improves positive-class probability ranking but does not guarantee simultaneous gains in all metrics at a frozen operating point. It matters especially for Test2, whose 46 toxic peptides mean that a small number of threshold-adjacent samples can substantially alter MCC and F1.

The benefit of the 3D-v2 expert was dataset-dependent. Test1 improved across the reported metrics, whereas on Test2 AUPRC increased but the fixed-threshold MCC decreased; these gains were measured under group-aware training OOF evaluation.

This result also clarifies the role of the structural expert within GHXTox. The 3D branch is most valuable as a source of complementary evidence rather than as a universal replacement for a strong sequence predictor, which motivates the final dual-expert probability fusion.

Table 6: Ablation analysis of the 3D structural expert. The first four rows use historical benchmark predictions; the final two rows are a paired five-fold OOF chemical-site experiment.

Evaluation	Variant	BACC	MCC	AUROC	AUPRC
Test1	Historical Cα 3D-v1	0.9179	0.8458	0.9699	0.9426
Test1	pLDDT-aware multi-scale 3D-v2	**0.9208**	**0.8464**	**0.9711**	**0.9474**
Test2	Historical Cα 3D-v1	**0.8481**	**0.6320**	**0.9316**	0.6969
Test2	pLDDT-aware multi-scale 3D-v2	0.8453	0.6128	0.9295	**0.7241**
Training OOF	Frozen backbone, no chemical-site residual	0.9153	0.8312	0.9687	0.9388
Training OOF	Frozen backbone + chemical-site residual	**0.9171**	**0.8342**	**0.9692**	**0.9401**

Figure 5: Module-level ablation results for the 1D and 3D experts (A–F).

3.5 Robustness, Statistical Uncertainty, and Applicability Analysis

Point estimates alone are insufficient for distinguishing a stable model improvement from variation caused by finite samples. We therefore use multi-seed averaging for the 1D expert and use paired, sample-level bootstrap resampling to compare the source-aware residual with its same-checkpoint zero-conformer OOF baseline. All comparisons preserve the pairing between the two predictions for each training peptide.

3.5.1 Multi-Seed Reproducibility

Three full-training 1D models initialized with seeds 42, 123, and 2025 are averaged before calibration. This ensemble is the final predictor; no seed is selected according to Test1 or Test2 performance. A 15-member ensemble of fold-specific checkpoints was also examined but not adopted because its members had each seen only approximately 80% of the training set and the aggregate did not improve the frozen validation criteria.

3.5.2 Paired Uncertainty and Point-Estimate Comparison

The current training-OOF comparison uses 5,000 class-stratified paired bootstrap resamples and 5,000 within-sample randomizations. Relative to the same-checkpoint zero-conformer OOF

vector, the MCC difference is +0.00190 (95% CI [−0.00043, 0.00454]; randomization $p = 0.203$), and the BACC difference is +0.00071 (95% CI [−0.00038, 0.00181]). Thus the point estimates favor the source-aware residual at the frozen threshold, but the intervals include zero.

The archived per-sample OOF vector for the immediately preceding pilot checkpoint is not available. Table 7 therefore reports deterministic benchmark differences between the source-aware full2795 model and that preceding pilot model, without attaching a full2795-versus-pilot confidence interval.

These retrospective differences are descriptive and were not used to tune the source-aware checkpoint.

Table 7: Point-estimate changes from the preceding pilot model to the source-aware full2795 model. Direct paired confidence intervals are not reported because the pilot per-sample OOF vector is unavailable.

Dataset	Metric	Current–previous Δ	Favorable direction
Test1	BACC	−0.00094	Increase
	MCC	−0.00018	Increase
	AUPRC	+0.00006	Increase
	Brier	−0.00030	Decrease
Test2	BACC	+0.00093	Increase
	MCC	+0.00725	Increase
	AUPRC	−0.00065	Increase
	Brier	−0.00082	Decrease

3.5.3 Controlled Multi-Conformer Ablation on Test1 and Test2

The contribution of the conformer pathway was isolated with a same-checkpoint ablation. The LoRA-ProtT5–Bi-Mamba probabilities, structural checkpoint, 0.72/0.28 fusion weights, and threshold of 0.47022256531284556 were held fixed; only the conformer statistics were either supplied or replaced by the exact availability-masked zero-conformer input. This design attributes the reported differences specifically to activation of the conformer residual, rather than to retraining, recalibration, or operating-point selection.

On Test1, activation of the source-aware residual changes one thresholded prediction, converting one false positive into a true negative. Consequently, SP, precision, BACC, F1, and MCC increase, whereas SN is unchanged. AUROC and AUPRC also increase slightly, and Brier score decreases by approximately 2.46×10^{-4}. The gains are directionally consistent but small in absolute magnitude.

On Test2, the source-aware residual also converts one false positive into a true negative. MCC increases by 0.00725, F1 by 0.00693, and Brier score decreases by approximately 6.44×10^{-4}, while SN is unchanged. AUROC is unchanged and AUPRC decreases by 2.62×10^{-4}. Across the two benchmarks, the conformer residual therefore corrected one false positive per set and reduced the Brier score on both, while Test2 AUPRC decreased slightly.

3.6 Interpretability and Representative Case Studies

All interpretation experiments used the frozen model and the prespecified case-selection rule described above. The analysis included 460 residue occlusions across 16 peptides, unperturbed confidence gates for all 1,708 benchmark records, 445 individual chemical-site deletions, 121 category-level deletions, and 424 residue-level site deletions. Before perturbation, baseline probabilities were recomputed and compared with the archived prediction vectors. The

Table 8: Controlled multi-conformer ablation on Test1. The zero-conformer and multi-conformer variants share the same trained checkpoint, 1D probabilities, fusion weights, and decision threshold. The multi-conformer cache covers 486 of 1,126 Test1 peptides. Δ is computed as multi-conformer minus zero-conformer; positive values are favorable for all listed metrics except Brier score, for which a negative value is favorable.

Metric	Zero conformer	Multi-conformer	Δ
ACC	0.947602	0.948490	+0.000888
BACC	0.932308	0.932929	+0.000620
Precision	0.916933	0.919872	+0.002939
SN	0.896875	0.896875	+0.000000
SP	0.967742	0.968983	+0.001241
F1	0.906793	0.908228	+0.001435
MCC	0.870460	0.872568	+0.002108
AUROC	0.981056	0.981099	+0.000043
AUPRC	0.967403	0.967482	+0.000080
Brier	0.040991	0.040745	−0.000246

Table 9: Controlled multi-conformer ablation on Test2. The zero-conformer and multi-conformer variants share the same trained checkpoint, 1D probabilities, fusion weights, and decision threshold. The multi-conformer cache covers 284 of 582 Test2 peptides through an exact unique-sequence join to Test1. Δ is computed as multi-conformer minus zero-conformer; positive values are favorable for all listed metrics except Brier score, for which a negative value is favorable.

Metric	Zero conformer	Multi-conformer	Δ
ACC	0.946735	0.948454	+0.001718
BACC	0.861778	0.862711	+0.000933
Precision	0.636364	0.648148	+0.011785
SN	0.760870	0.760870	+0.000000
SP	0.962687	0.964552	+0.001866
F1	0.693069	0.700000	+0.006931
MCC	0.667340	0.674587	+0.007248
AUROC	0.943300	0.943300	+0.000000
AUPRC	0.804590	0.804328	−0.000262
Brier	0.043631	0.042987	−0.000644

maximum absolute discrepancy for the 1D expert within the case cohort was 3.45×10^{-4}; the corresponding 3D discrepancies were 5.49×10^{-4} on Test1 and 1.14×10^{-5} on Test2. None of these differences changed a standalone 3D or fused decision. The perturbation baselines therefore preserved the archived decision states, with the larger Test1 discrepancy reported as bounded cross-environment numerical drift.

3.6.1 Residue-Level Interpretation of the 1D Expert

The residue-occlusion profiles were heterogeneous across peptides and seeds. Across the 16 prespecified cases, the mean pairwise Spearman correlation of absolute per-residue logit effects was 0.118 (case-level range, −0.077 to 0.422), whereas the mean residue-wise sign-agreement fraction was 0.740 (range, 0.667–0.796). Direction was therefore more stable across seeds than the fine-grained ranking of residue magnitudes. The reserved maps emphasize local effects with consistent signs rather than implying a single stable ranking across all residues.

Several cases nevertheless contained large, directionally consistent effects. For the Test2 toxic peptide `sequence_34_34`, occlusion of Asn9 reduced the calibrated 1D probability from 0.5447 to 0.0412 ($\Delta p_{1D} = +0.5035$), with all three seeds agreeing on the sign. In the Test1 toxic case `peptide_199`, occlusion of Ser24 reduced the score from 0.3687 to 0.1114 ($\Delta p_{1D} = +0.2573$), again with complete sign agreement. Effects were not uniformly toxicity-supporting: in the Test2 toxic case `sequence_23_23`, occlusion of Ile5 increased the 1D probability from 0.2807 to 0.4819 ($\Delta p_{1D} = -0.2013$).

These cases illustrate large, context-dependent sensitivities in the 1D output; they do not identify experimentally validated toxic residues.

The dataset-specific layouts are reserved in Figures 6 and 7, respectively.

Reserved for the Test1 residue-occlusion figure

Eight sequence heat strips will show one toxic and one non-toxic peptide from each expert-outcome group. A shared signed color scale will encode within-peptide L_1-normalized Δp_{1D}; residue positions, seed sign agreement, baseline probability, and expert-outcome group will be annotated. Cases are ordered by the prespecified cohort rule, not by attribution magnitude.

Figure 6: Reserved layout for Test1 residue-level analysis of the frozen 1D expert. Each row will represent one independently selected peptide and each cell one residue. Positive values will indicate that zeroing the post-contextual residue vector lowers the calibrated 1D toxicity probability, whereas negative values will indicate an increase. The common color normalization will permit comparison of direction and relative within-peptide magnitude; it will not imply that equal colors correspond to equal biological effects across peptides. Seed agreement will distinguish reproducible directions from effects dominated by one checkpoint.

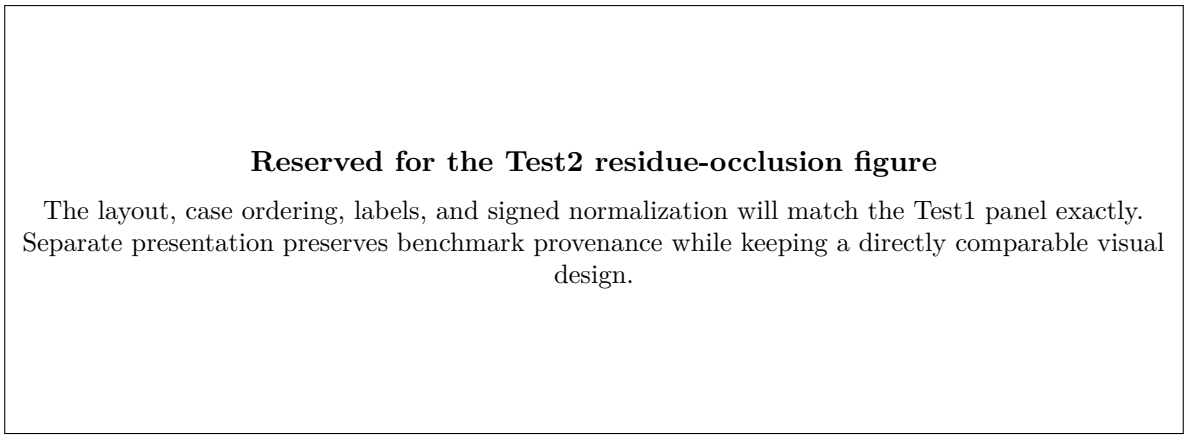


Figure 7: Reserved layout for Test2 residue-level analysis of the frozen 1D expert. The eight rows will cover both labels in all four expert-outcome groups. Residue-wise probability shifts will be computed with the three frozen seeds, the original inference padding context, and the fixed OOF Platt transform. Test2 comprises exact sequence matches to a subset of Test1, so matched records may yield similar maps. Such agreement will be treated as a reproducibility check, not as independent external validation.

3.6.2 Structural Confidence and Chemical-Site Sensitivity

Mean pLDDT was strongly associated with the learned global structural gate. Spearman correlations were 0.960 on Test1 (95% peptide-bootstrap interval, 0.953–0.966) and 0.950 on Test2 (0.937–0.959); the pooled descriptive value was 0.957 (0.951–0.961). In contrast, the mean within-peptide correlation between residue pLDDT and node gates was small: 0.049 on Test1 (0.030–0.068) and 0.064 on Test2 (0.036–0.092). Thus, global gating closely tracked confidence at the peptide level, whereas the weak local correlations are consistent with residue gates also reflecting geometric and contextual features.

Most individual-site deletions produced small changes, but the distribution had a pronounced upper tail. Across 445 deletions, the median and mean absolute changes in p_{3D} were 0.00048 and 0.00997, respectively; the 95th percentile was 0.0570 and the maximum was 0.2994. Nine individual deletions moved the recomputed fused probability across the frozen decision threshold when the 1D probability was held fixed. At category level, hydrophobic-site deletion had the largest mean absolute change among the eight modeled categories (0.126 across 16 case-level deletions), followed by aromatic sites (0.0687), whereas sulfur-site deletion had the smallest mean absolute change (0.00179). Because the analysis used the selected 16-peptide cohort, these category-level estimates are descriptive, not population effects.

The largest individual-site effect occurred at the aromatic/hydrophobic Phe8 site in the matched cache-covered cases `peptide_846` and `sequence_388_388`: deletion increased p_{3D} from approximately 0.2405 to 0.5399, giving $\Delta p_{3D} \approx -0.2994$. Conversely, deletion of the hydrophobic/conformational Pro19 site in `peptide_199` decreased p_{3D} from 0.3443 to 0.1163 ($\Delta p_{3D} = +0.2280$).

The opposing directions show that the deletion score depends on the surrounding peptide context and cannot be read as an intrinsic toxicity label for a functional group.

Figure 8 reserves a layout for these scale-specific associations and perturbation effects.

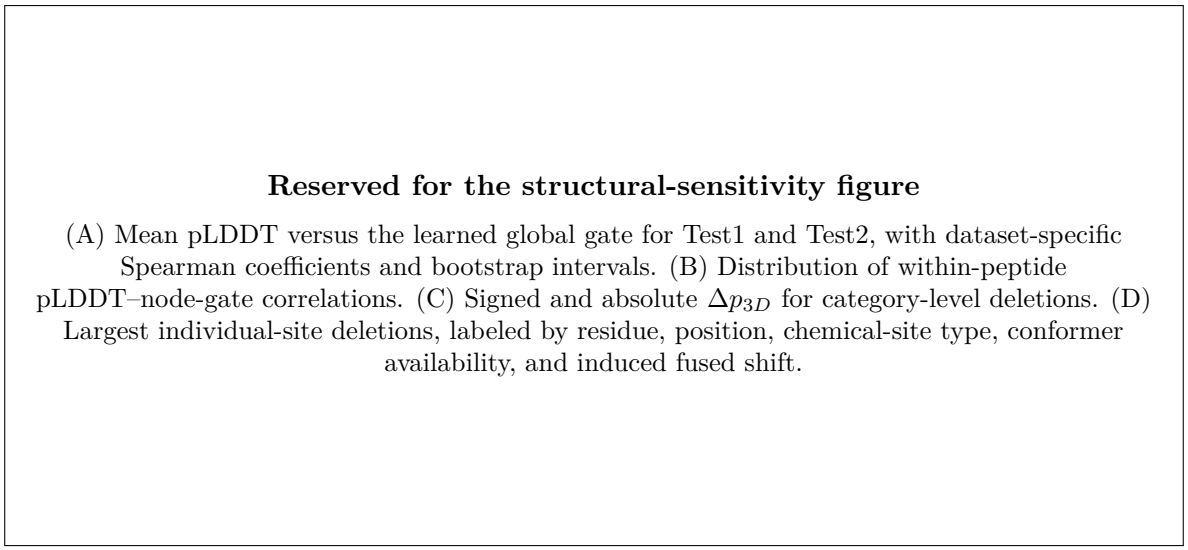


Figure 8: Reserved layout for structural-confidence and chemical-site sensitivity analysis. (A) Association between mean pLDDT and the global gate. (B) Distribution of within-peptide pLDDT–node-gate correlations. (C) Comparison of the eight modeled chemical categories. (D) Largest probability-increasing and probability-decreasing site-deletion effects. The deletion panels use the frozen 3D expert and calculate Δp_{fused} with p_{1D} fixed; the reported values denote model sensitivity, not experimentally verified toxicophores.

3.6.3 Expert Agreement, Disagreement, and Fusion Outcomes

The two experts were both correct for 1,032 of 1,126 Test1 peptides (91.65%) and 529 of 582 Test2 peptides (90.89%). On Test1, the 1D expert alone was correct for 29 cases (2.58%), the 3D expert alone for 23 (2.04%), and both were incorrect for 42 (3.73%). The corresponding Test2 counts were 13 (2.23%), 14 (2.41%), and 26 (4.47%). The median absolute expert-probability gaps were small in the both-correct groups (0.0145 on Test1 and 0.0126 on Test2) but exceeded 0.49 in each one-expert-correct group. The latter cases therefore reflect substantial disagreement, not threshold-scale numerical variation.

Among samples for which only one expert was correct, the fixed 0.72/0.28 fusion remained correct in 19 of 29 1D-only-correct Test1 cases and 17 of 23 3D-only-correct cases. The corresponding Test2 counts were 10 of 13 and 12 of 14.

For example, the toxic Test2 peptide `sequence_23_23` had $p_{1D} = 0.2806$, $p_{3D} = 0.9954$, and $p_{\text{fused}} = 0.4808$; the high 3D score moved the fused prediction above the decision threshold. By contrast, the toxic `sequence_34_34` had $p_{1D} = 0.5447$ but $p_{3D} = 3.0 \times 10^{-5}$, yielding $p_{\text{fused}} = 0.3922$ and a false negative despite the correct standalone 1D decision.

These contrasting cases show that structural evidence can either correct or introduce an error under a fixed weighted fusion.

Figure 9 reserves a common layout for the outcome counts, probability disagreements, and cache-covered case studies.

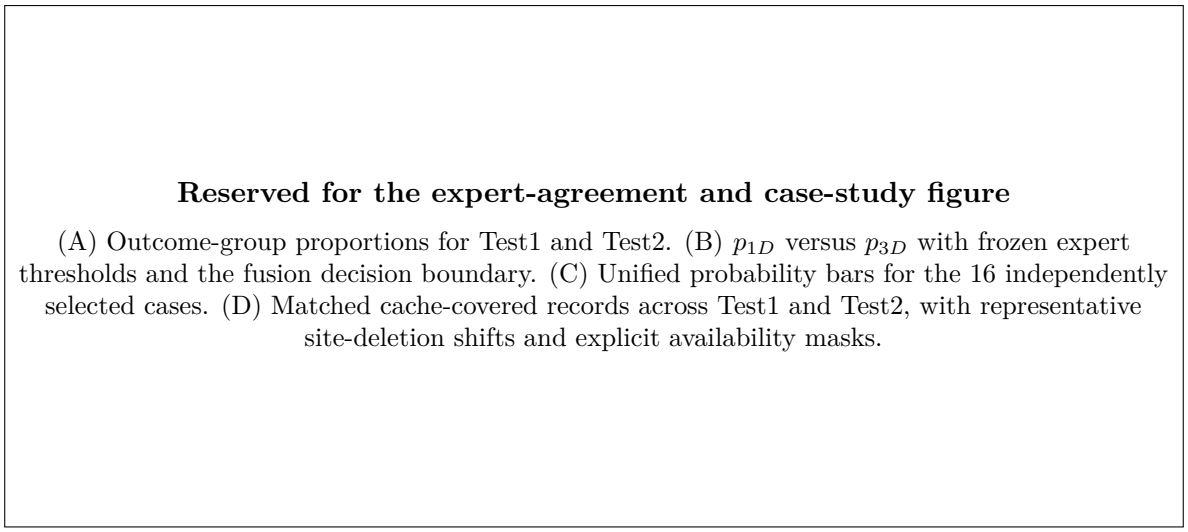


Figure 9: Reserved layout for expert agreement, disagreement, and representative cases. (A) Proportions of all four predeclared outcome groups. (B) Magnitude and direction of expert disagreement. (C) Probability profiles for the label-balanced, threshold-proximal perturbation cohort, annotated by whether fusion agrees with the individually correct expert. (D) Matched, cache-covered cases shown as sensitivity analyses of predicted-structure inputs, not as experimental conformers or molecular-dynamics trajectories.

3.6.4 Representative Cases with Conformer Coverage

The interpretability cohort contains cases with conformer-cache coverage and cases assigned the prespecified zero-residual fallback. Among the 445 individual-site perturbations, 144 belonged to cache-covered peptides and 301 to fallback peptides. Their mean absolute Δp_{3D} values were 0.0166 and 0.00678, respectively. This unmatched comparison cannot isolate an effect of conformer availability because the cases differ in sequence, label, length, site count, and baseline confidence.

Several selected Test2 peptides exactly match sequences selected from Test1 and therefore provide a cross-dataset reproducibility check for preprocessing and perturbation. For the cache-covered pair `peptide_481`/ `sequence_143_143`, deleting the donor/acceptor Asn18 site changed p_{3D} from approximately 0.4653 to 0.6420 in both benchmark records ($\Delta p_{3D} \approx -0.1767$), and deleting the hydrophobic Ile6 site gave $\Delta p_{3D} \approx -0.1431$ in both. The near-identical values are consistent with the declared sequence-based reuse of the Test2 cache and the deterministic perturbation procedure. They are not independent replicates because Test2 comprises exact sequence matches to a subset of Test1. No conclusions about conformational dynamics are drawn because aligned trajectories, RMSF, contact occupancy, and experimentally determined conformational transitions were not analyzed.

3.7 Limitations and Future Directions

Several limitations remain. Test1 and Test2 were observed during earlier model development, so they are frozen historical benchmarks rather than blinded external test sets; structural predictions remain uncertain; and conformer coverage is incomplete.

3.7.1 Dataset and Evaluation Limitations

Test2 contains only 46 toxic peptides, so its SN, BACC, and MCC are sensitive to a few prediction changes. Stronger generalization evidence will require a new dataset evaluated under a prefrozen protocol.

3.7.2 Structural and Conformer Limitations

pLDDT reflects predicted confidence rather than experimental accuracy or conformational flexibility, and full-atom-derived chemical sites are candidate spatial-chemical relationships rather than confirmed bonds. Conformer coverage is 43.16% of Test1 and 48.80% of Test2, with generator identity partly coupled to peptide length; the conformer pathway is therefore a conditional refinement, not a universal source of improvement. More flexible residual variants did not yield a stable multimetric improvement, and the single sample-level toxicity label does not specify which conformer or microstate should carry the activity.

3.7.3 Modeling and Interpretability Limitations

The two experts are trained separately and combined with a fixed fusion weight that cannot adapt per sample. Interpretability analyses are limited in scope: residue occlusion covers a 16-peptide cohort with modest cross-seed rank agreement, and chemical-site deletion operates on predicted-structure features.

3.7.4 Future Directions

Future work should prioritize a new independent external dataset, better-controlled same-sample conformer sets, and representation learning for the conformer pathway rather than further fixed aggregators. Orthogonal attribution methods and, where feasible, mutational or toxicity experiments could then test the resulting hypotheses.

4 CONCLUSIONS

GHXTox combines a task-adaptive LoRA-ProtT5–Bi-Mamba sequence expert with a pLDDT-aware multi-scale structural expert through a fixed probability fusion protocol. The structural expert integrates residue geometry, full-atom-derived chemical sites, and an availability-masked conformer residual, and the final predictor uses the structural expert's standalone probability. The 1D/3D weights (0.72/0.28) and the decision threshold (0.47022256531284556) were fixed from training OOF predictions before benchmark evaluation.

On the frozen historical Test1 and Test2 benchmarks, the model reaches MCC values of 0.8726 and 0.6746 and AUPRC values of 0.9675 and 0.8043, and the final fusion improves AUROC, AUPRC, and the Brier score relative to the calibrated 1D stream. The conformer residual corrects one false positive per benchmark and reduces the Brier score on both, while Test2 AUPRC decreases slightly; its training-OOF MCC gain (+0.00190, 95% CI [−0.00043, 0.00454]) is not statistically established. Marked conformational variation coexisted with nearly invariant frozen-expert probabilities, indicating a bottleneck in mapping conformational variation to toxicity labels. The interpretability analyses describe model sensitivity rather than causal mechanisms.

Several limitations remain: both benchmarks were observed during earlier model development, the structures are predicted rather than experimental, and conformer coverage is incomplete. Future work should prioritize a new independent external dataset, better-controlled same-sample conformer sets, and representation learning for the conformer pathway, followed where feasible by experimental validation.

Data and Software Availability

The source code, configuration files, fold manifests, and inference scripts used in this study are available at `https://github.com/Suiyumanbu/GHXTox`. A versioned archival release containing the exact commit identifier, environment manifest, frozen model parameters, calibration

coefficients, fusion weights, full-precision decision threshold, and sample-level prediction files will accompany the final publication. Raw peptide records were obtained from the public databases described in Section 2.1. Where redistribution is restricted by the original database terms, accession lists and preprocessing scripts are provided instead of redistributing the source records.

Supporting Information

A Supporting Information package containing detailed dataset documentation, fold-wise results, hyperparameter settings, additional ablation experiments, subgroup analyses, statistical outputs, and extended interpretability case studies will accompany the submitted manuscript.

CRediT Authorship Contribution Statement

Acknowledgements

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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